Random Centres and the ε-Dependence of Constant-Round Planar Domination

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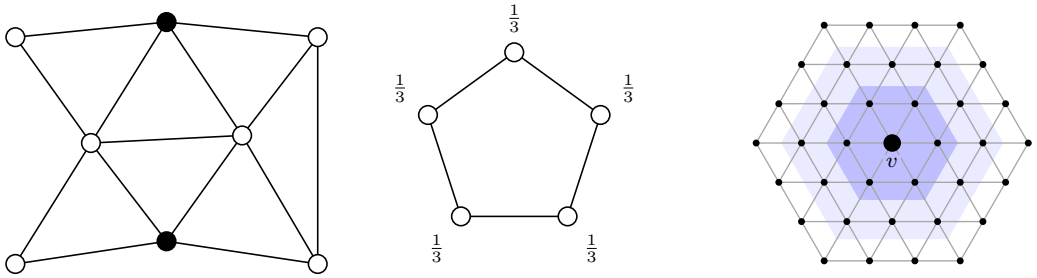
Abstract

It was recently shown (arXiv:2610.09618) that the infimum of the approximation ratios of deterministic constant-round LOCAL algorithms for minimum dominating set on planar graphs is 7. The $(7+\varepsilon)$-approximation given there runs in $T_0 + O((c_0/\varepsilon)^\kappa)$ rounds with $\kappa = \ln 3/\ln(24/23) \approx 25.81$, and it is asked there whether this exponent can be substantially reduced. We replace the heaviest-edge star selection in its clustering step by a random-centre rule. Every part becomes a centre independently with probability p, and every non-centre joins its heaviest centre neighbour. For planar graphs, a layer-cake count combined with two degree inequalities shows that for $p \geq 21/64$ each level contracts at least an expected $(1-p)\big(3-2(1-p)^3-(1-p)^6\big)/9$ fraction of the remaining weight. With $p = 3/8$, which takes three fair coins per part and level, this gives $\kappa < 5.89$. With $p = 1/2$, that is, one fair coin per part and level as in the original algorithm, it gives $\kappa < 6.7$. An analysis that counts only these join edges cannot give an exponent below 5.74. For every proper minor-closed class of density a, a bounded out-degree orientation gives the fraction $(1-p)(1-(1-p)^k)/k$ with $k = \lceil a \rceil$. The new exponent is never worse, and with about $\frac{1}{2}\log_2 a$ fair coins per part and level it is smaller by a factor that tends to 8 as the density grows. This is optimal for the rule when a is an integer and the class contains all $K_{a,n}$. A bipartite density bound or a non-integer density improves it further. In particular, the planar bounds hold verbatim on the torus, the projective plane and the Klein bottle. We then turn to the dependence on ε of the $(1+\varepsilon)$-approximations. On paths, every deterministic r-round algorithm for fractional domination, and every randomized one that always outputs a dominating set, has (expected) ratio at least $1 + 1/(r+4)$. So these approximations need $\Omega(1/\varepsilon)$ rounds on every minor-closed class that contains all paths. For fractional domination on paths of known length, this is tight up to an additive constant. On forests and on outerplanar graphs, with no bound on the degree, $O(1/\varepsilon)$ rounds suffice for both approximations, so there the optimal number of rounds is $\Theta(1/\varepsilon)$. Machine-checked in Lean 4 with Mathlib are the level bounds for every $p \in [21/64, 1]$, assuming only the classical edge bounds of planar graphs; the lower bound on paths, and the upper bound for algorithms that know the length, in an explicit model of r-round algorithms; the deterministic upper bound on forests and the size bound of the randomized one; and the limits of the approach.

1 Introduction

1.1 Dominating sets and the LOCAL model

A set D of vertices of a graph G is *dominating* if every vertex lies in D or has a neighbour in D. The *domination number* $\gamma(G)$ is the minimum size of a dominating set, and minimum dominating set (MDS) is the problem of finding one. The integer program $\min \sum_v x_v$ subject to $\sum_{u \in N[v]} x_u \geq 1$ for every v, where $N[v]$ is the closed neighbourhood of v and $x_v \in \{0,1\}$, has optimum $\gamma(G)$. Relaxing it to $x_v \geq 0$ gives the *fractional domination number* $\gamma_f(G) \leq \gamma(G)$ (Figure 1).



(a) a dominating set D (black), $\gamma = 2$

(b) $x_v = \frac{1}{3}$ is optimal: $\gamma_f(C_5) = \frac{5}{3} < 2 = \gamma(C_5)$

(c) what v knows after $T = 1$ and $T = 2$ rounds

Figure 1: Dominating sets and local views. In (c), the shaded hexagons are the radius-1 and radius-2 neighbourhoods of v, from which a 1-round and a 2-round LOCAL algorithm decide the output of v.

MDS is NP-hard, even on planar graphs of maximum degree 3 [7]. It is a set cover problem whose sets are the closed neighbourhoods. The greedy algorithm, which repeatedly adds a vertex that dominates the most undominated vertices, therefore achieves ratio $1 + \ln(\Delta + 1)$ on graphs of maximum degree Δ [12], and by the standard reduction from set cover, ratio $(1 - \varepsilon) \ln n$ is NP-hard [6]. Planar graphs are much easier: the layering technique of [2] gives a polynomial-time $(1 + \varepsilon)$-approximation for every fixed $\varepsilon > 0$.

In distributed computing the input graph is the network itself. Every vertex is a processor that initially knows only its identifier and its incident edges, and it decides whether it belongs to D by exchanging messages with its neighbours. Dominating sets arise in this form as cluster heads in wireless ad hoc and sensor networks, where every node needs a coordinator within one hop and no node knows the whole topology. In the LOCAL model [18], computation proceeds in synchronous rounds. Vertices have distinct identifiers from a finite range, and in a randomized algorithm every vertex has its own infinite sequence of independent fair coins. In every round each vertex sends a message of arbitrary size to each neighbour and performs arbitrary local computation. After T rounds a vertex knows exactly its radius-T neighbourhood, with the identifiers and the coins in it, so a T-round algorithm is a function from radius-T neighbourhoods to outputs, and the number of rounds measures how far a vertex has to look. A constant-round algorithm is local in the strongest sense. Its running time does not depend on the size of the network, a change in the graph affects only the outputs within constant distance, and on graphs of bounded degree it yields a sublinear-time algorithm that decides the output of a single vertex by exploring its neighbourhood [22].

1.2 Standard distributed algorithms

Parallel greedy. The greedy algorithm adds one vertex at a time. Its direct distributed version lets a vertex join when its *span*, the number of undominated vertices in its closed neighbourhood, is the largest within distance 2, so that vertices joining in the same round do not compete for the same undominated vertices. This can still take $\Omega(n)$ rounds, for example on a caterpillar whose spine vertices have decreasing spans, because then only one vertex joins per round. The algorithm of [11] removes this bottleneck by rounding spans up to powers of 2, so that many vertices tie. In every round, the vertices whose rounded span is maximal within distance 2 become candidates, every undominated vertex counts the candidates that dominate it, and every candidate v joins with probability $1/\mathrm{med}(v)$, where $\mathrm{med}(v)$ is the median of these counts over the undominated vertices

that v dominates. The expected size of the result is $O(\log \Delta)\gamma(G)$, and the algorithm terminates in $O(\log n \log \Delta)$ rounds with high probability.

LP relaxation and rounding. A fractional dominating set x becomes a dominating set in two rounds. Every vertex joins independently with probability $\min\{1, x_v \ln(\Delta+1)\}$, and every vertex that is still undominated joins as well. Since $\sum_{u \in N[v]} x_u \geq 1$, a vertex remains undominated after the first step with probability at most $e^{-\ln(\Delta+1)} = 1/(\Delta+1)$. Every vertex dominates at most $\Delta+1$ vertices, so $n/(\Delta+1) \leq \gamma_f(G)$, and the expected size is at most $\ln(\Delta+1)\sum_v x_v + \gamma_f(G)$. In [15], covering and packing LPs are solved within a factor $1+\varepsilon$ in $O(\log n/\varepsilon)$ rounds. Together with a rounding of this kind, this gives an $O(\log \Delta)$-approximation of MDS in $O(\log n)$ rounds. In the same paper it was shown that every polylogarithmic approximation, even a randomized one, needs $\Omega\big(\min\{\log \Delta/\log\log \Delta, \sqrt{\log n/\log\log n}\}\big)$ rounds. So on general graphs no constant-round algorithm has a constant ratio.

Planar graphs in a constant number of rounds. Sparse classes behave differently. The algorithm of [16] has two steps. First, every vertex v whose open neighbourhood cannot be dominated by six vertices other than v joins D. This needs only the radius-2 neighbourhood of v. Second, every vertex that is still undominated adds a vertex of its closed neighbourhood that dominates the most undominated vertices. The algorithm runs in six rounds and has ratio 130 on planar graphs [16]. Planarity enters only in the analysis, which shows that the first step selects fewer than $3\gamma(G)$ vertices outside a minimum dominating set. This raised the question of the best ratio that a constant number of rounds can achieve on planar graphs (Section 1.3).

Cluster-and-solve. Ratios close to the optimum need a global view of larger regions. Algorithms of this kind on sparse classes [5, 17] partition the graph into clusters of small radius such that few edges run between clusters, let one vertex of every cluster collect the cluster and solve it exactly, and combine the local solutions. The quality of the result depends on the weight of the edges between clusters, and the number of rounds on the cluster radius. In [5], this gives a $(1+\varepsilon)$-approximation of MDS on planar graphs in $O(\log^* n)$ rounds. A common way to build the clusters is repeated star contraction: every cluster picks a neighbouring cluster to merge with, such that the merges form vertex-disjoint stars. Every star becomes one cluster, and parts of radius at most r give a cluster of radius at most $3r+1$ (Figure 2).

1.3 Optimal ratio on planar graphs

In [10] it was shown that no deterministic constant-round algorithm achieves ratio $7-\varepsilon$ on planar graphs. A sequence of works improved the upper bound [9, 23], and the gap was recently closed in [1]: for every $\varepsilon > 0$ there is a deterministic algorithm that outputs a dominating set of size at most $(7+\varepsilon)\gamma_f(G)$ in

$$T_0 + O\big((c_0/\varepsilon)^\kappa\big) \text{ rounds}, \qquad \kappa = \frac{\ln 3}{\ln(1/q)}, \quad q = 1 - \frac{1}{8a} = \frac{23}{24},$$

where $a = 3$ is the edge density of planar graphs (every subgraph H has at most $a|V(H)|$ edges) and T_0, c_0 are absolute constants. The exponent $\kappa \approx 25.81$ comes from a randomized star-contraction clustering [17, 5]: each level grows part radii by a factor of 3 and reduces the expected weight between parts by a factor q. Bringing this weight down to an ε-fraction takes $L \approx \ln(1/\varepsilon)/\ln(1/q)$ levels, and hence radius and round number $3^L \approx (1/\varepsilon)^\kappa$. Open Problem 4 of [1] asks whether this exponent can be substantially reduced.

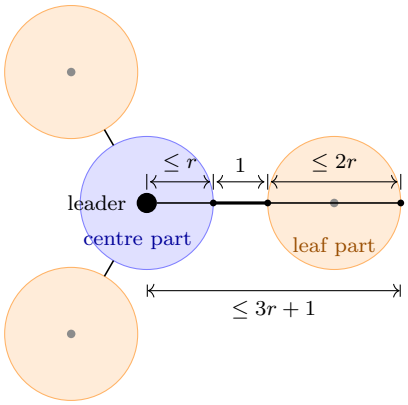


Figure 2: A star of parts of radius at most r becomes one part of radius at most $3r+1$ around the leader of its centre part.

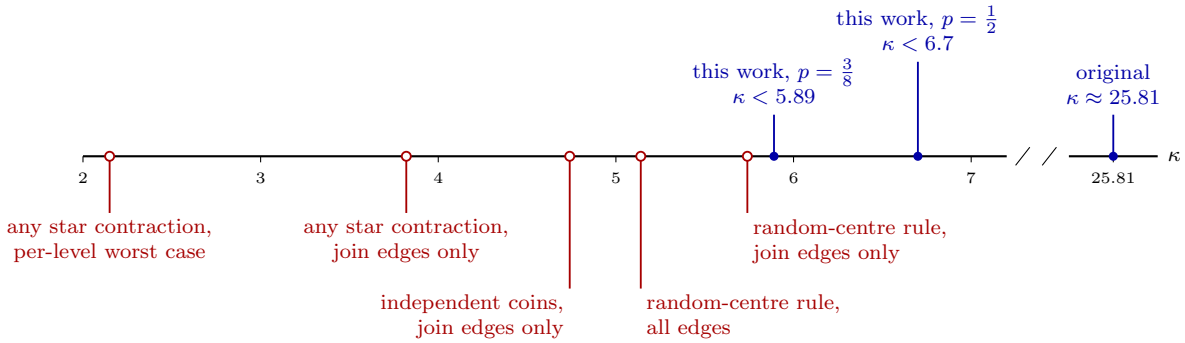


Figure 3: Round exponents for planar graphs. Above the axis: proved upper bounds on κ. Below the axis: the smallest exponent that the indicated kind of analysis can give (Section 4).

Our result. We change only the star-selection rule of the clustering step (Section 2). For planar graphs and $p \in [21/64, 1]$, the per-level reduction factor improves from $23/24$ to $1-\varphi(p)$ with $\varphi(p) = (1-p)\big(3-2(1-p)^3-(1-p)^6\big)/9$ (Theorem 10). For a class of density a, it improves from $1-1/(8a)$ to $1-f_k(p)$ with $f_k(p) = (1-p)(1-(1-p)^k)/k$ and $k = \lceil a \rceil$ (Theorem 4). Everything else in [1] carries over unchanged (Section 3): the radius recursion, the view radius, the derandomization and the rounding.

Theorem 1. *For every $\varepsilon \in (0,1]$ there is a deterministic LOCAL algorithm that returns, on every planar graph G, a dominating set of size at most $(7+\varepsilon)\gamma_f(G)$ in $T_0 + O((c_0/\varepsilon)^{\kappa'})$ rounds, where $\kappa' = \ln 3/\ln(2097152/1740037) < 5.89$, and T_0, c_0 are as in [1, Cor. 8.1]. With one fair coin per part and level, as in [1], the same holds with $\kappa' = \ln 3/\ln(1152/977) < 6.7$, and with six fair coins ($p = 21/64$) it holds with $\kappa' < 5.83$.*

The first bound uses three fair coins per part and level ($p = 3/8$). The six-coin exponent is within 10^{-3} of the best exponent that Theorem 10 gives for any p. Figure 3 compares the exponents for $p = 3/8$ and $p = 1/2$ with the limits of Section 4.

For a general proper minor-closed class of density a, the exponent $\ln 3/\ln(1/q)$ improves as in Table 1. Lemma 5 shows that the new factor is never worse.

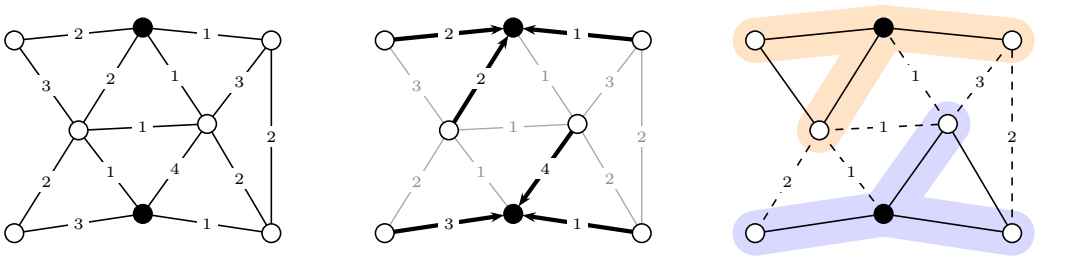
a	[1]: $q = 1 - \frac{1}{8a}$	ours, $p = \frac{1}{2}$	ours, best dyadic p	improvement
1	8.23	3.82	3.82	2.15×
2	17.02	5.30	5.14	3.31×
3 (planar)	25.81	6.97	6.42	4.02×
6	52.18	12.84	10.08	5.17×
10	87.34	21.44	14.81	5.89×
100	878.34	219.18	115.67	7.59×

Table 1: Round exponent $\kappa = \ln 3 / \ln(1/q)$ for integer density a (so $k = a$). The best dyadic p uses at most 8 fair coins per part and level. Our exponents are rounded up, the improvement factors down. For planar graphs, Theorem 10 improves the row $a = 3$ to 6.67 ($p = 1/2$) and 5.83 ($p = 21/64$).

Positioning. The improvement concerns the optimal-ratio regime. The algorithm of [23] is a deterministic $(8+\varepsilon)$-approximation whose last phase is that of [9]. That phase reduces to maximum degree $\Gamma = O(1/\varepsilon)$ by selecting vertices at a constant-factor cost, and then runs the LP algorithm of [15] in $O(r^2)$ rounds with $r = \log \Gamma / \log(1+\varepsilon)$, roughly $(\log(1/\varepsilon)/\varepsilon)^2$. That degree reduction loses a constant factor. To reach ratio 7, the algorithm of [1] instead computes a $(1+\varepsilon)$-approximation of γ_f on the whole graph by cluster-and-solve. Among algorithms whose number of rounds does not depend on n, it is the only one known for target ratios in $(7, 8]$, and Theorem 1 is its fastest version. The same holds for the other guarantees of [1, Thm. 6.2] that our substitution improves for every proper minor-closed class: the deterministic $(1+\varepsilon)$-approximation of γ_f, the randomized algorithm with expected size $(1+\varepsilon)\gamma$, and the ratio $2a + 1 + \varepsilon$. On forests and on outerplanar graphs, Theorems 19 and 20 give much faster algorithms for the first two. Reaching ratio $7 + \varepsilon$ in $\mathrm{poly}(1/\varepsilon)$ rounds with an exponent close to 2 remains open (Section 7).

The dependence on ε. Open Problem 3 of [1] asks for the optimal dependence on ε of the deterministic $(1+\varepsilon)$-approximation of γ_f and of the randomized algorithm with expected size $(1+\varepsilon)\gamma$. On paths, both need at least $1/\varepsilon - 4$ rounds (Theorem 17), so $\Omega(1/\varepsilon)$ rounds are necessary on every minor-closed class that contains all paths. On forests (Theorem 19) and on outerplanar graphs of arbitrary maximum degree (Theorem 20), both run in $O(1/\varepsilon)$ rounds, so the answer there is $\Theta(1/\varepsilon)$. The algorithms average the optimal solutions of overlapping windows of radius $O(1/\varepsilon)$. On outerplanar graphs the windows live in a covering graph in which every long face unrolls into an infinite path.

Formal verification. The bounds for one level of the rule (Section 2), in the planar case for every $p \in [21/64, 1]$ and on the surfaces of Table 2, the resulting exponents and their limit as the density grows (Proposition 6), the entries of Table 1, Proposition 14 and Theorem 17 are checked in Lean 4 with Mathlib (Section 6). The direction of the orientation theorem of [8] that we use and Lemmas 8 and 11 are checked as well. Hence the level bounds assume only the classical edge bounds of planar graphs from Euler's formula: at most $3n$ edges, and at most $2n$ if the graph is bipartite. Theorem 17 is checked in a model of r-round algorithms on paths, for deterministic and for randomized algorithms, and so is the upper bound $1 + 2/(2r+3)$ on paths for algorithms that know the length. The integration into the algorithm of [1] is proved on paper, except for the step from the per-level factor to the exponent. The deterministic part of Theorem 19 is checked as well, with its explicit constants. For its randomized part, Lean checks the expected size, the



(a) parts whose coins come up heads become centres (black)

(b) every other part joins its heaviest centre neighbour

(c) two new parts; the weight between parts drops from 28 to 10

Figure 4: One level of the random-centre rule on a quotient Q of total weight 28. The join edges (bold in (b)) carry weight 13. The two edges between leaves of the same star, of weights 2 and 3, become internal as well.

cut probability and the dependence on the random values, with the input of [16] as a hypothesis; that the clusters are computed locally is proved on paper. Theorem 20 is proved on paper, and we check its construction numerically. Lean checks the steps that do not depend on the graph class (the packing bound, the repair and the averaging over windows), the run lemmas, the interval facts of the randomized version and the arithmetic of all constants, with the structure of the covering graph as hypotheses. The limits of Section 4 are checked as well, with the planarity of the graphs used there taken as given.

2 One level of random-centre contraction

Let Q be a finite simple graph with edge weights $w\colon E(Q) \to \mathbb{R}_{\geq 0}$ and distinct identifiers. For $C \subseteq V(Q)$, a vertex $P \notin C$ with a neighbour in C *joins* a neighbour $c(P) \in C$ that maximizes $w(Pc(P))$, breaking ties by identifier. The *join edges* are the edges $Pc(P)$, and the *gain* is $\mathrm{gain}(C) = \sum_{P \notin C} \max_{X \in C \cap N(P)} w(PX)$, with $\max \emptyset = 0$. Figure 4 shows an example.

Lemma 2. *Distinct joining vertices have distinct join edges, so* $\mathrm{gain}(C)$ *is the total weight of a set of edges. Contracting each centre together with the vertices that joined it yields vertex-disjoint stars, and every join edge becomes internal.*

Proof. A join edge $Pc(P)$ has exactly one endpoint outside C, namely P. Hence $Pc(P) = P'c(P')$ forces $P = P'$. Centres do not join, and every non-centre joins at most one centre. □

For a finite set A let $C_A \subseteq A$ contain each element independently with probability p. Write $s = 1 - p$ and $g(m) = 1 - s^m$.

Lemma 3 (vertex lemma)**.** *For $p \in [0,1]$ and weights $w\colon A \to \mathbb{R}_{\geq 0}$,*

$$|A| \cdot \mathbb{E}\Big[\max_{x \in C_A} w(x)\Big] \;\geq\; g(|A|) \sum_{x \in A} w(x).$$

Proof. We use induction on $n + 1 = |A|$; the case $A = \emptyset$ is trivial. Let $y \in A$ have maximum weight, and put $A' = A \setminus \{y\}$, $W' = \sum_{x \in A'} w(x)$ and $E' = \mathbb{E}[\max_{x \in C_{A'}} w(x)]$. Conditioning on whether $y \in C_A$ gives $\mathbb{E}[\max_{C_A} w] = sE' + pw(y)$, because y is the heaviest centre whenever it is a centre. The induction hypothesis gives $nE' \geq g(n)W'$, and the choice of y gives $W' \leq nw(y)$. Using $g(n+1) = 1 - s\,s^n$, a direct expansion verifies

$$n\Big[(n{+}1)\big(sE'{+}pw(y)\big) - g(n{+}1)\big(w(y){+}W'\big)\Big] = (n{+}1)s\big(nE' - g(n)W'\big) + \big(nw(y) - W'\big)\big(np - s\,g(n)\big).$$

Both terms are non-negative. For the second, Bernoulli's inequality gives $g(n) \le np$, and $s \le 1$. For $n \ge 1$ we divide by n. For $n = 0$ we have $W' = 0$, and the claim reads $pw(y) \le sE' + pw(y)$. □

Since g is concave with $g(0) = 0$, the ratio $g(m)/m$ is non-increasing. Hence $g(m)/m \ge g(k)/k$ for $1 \le m \le k$.

Theorem 4. *Suppose the edges of Q admit an orientation in which every vertex has out-degree at most $k \ge 1$. If every vertex of Q is in C independently with probability $p \in [0,1]$, then*

$$\mathbb{E}[\mathrm{gain}(C)] \;\ge\; f_k(p)\, w(Q), \qquad f_k(p) = \frac{(1-p)\big(1-(1-p)^k\big)}{k}.$$

Proof. Let $O(P)$ be the out-neighbourhood of P, so $w(Q) = \sum_P \sum_{X \in O(P)} w(PX)$. Fix P. Then $\max_{X \in C \cap N(P)} w(PX) \ge \max_{X \in C \cap O(P)} w(PX)$. The event $P \notin C$ is independent of $C \cap O(P)$, because $P \notin O(P)$. If $O(P) = \emptyset$, the sum below is empty. Otherwise, by Lemma 3 applied to $A = O(P)$ and by the monotonicity of $g(m)/m$,

$$\mathbb{E}\Big[[P \notin C] \max_{X \in C \cap N(P)} w(PX)\Big] \;\ge\; (1-p)\frac{g(|O(P)|)}{|O(P)|} \sum_{X \in O(P)} w(PX) \;\ge\; f_k(p) \sum_{X \in O(P)} w(PX).$$

Summing over P proves the claim. □

By [8], a graph has an orientation with out-degree at most k if and only if every subgraph H satisfies $|E(H)| \le k|V(H)|$. A graph in a class of density a therefore admits such an orientation with $k = \lceil a \rceil$. For planar graphs $k = 3$. Over $p \in [0,1]$, f_3 is maximized at $p = 1 - 4^{-1/3} \approx 0.370$. The dyadic choice $p = 3/8$ (three fair coins) gives $f_3(3/8) = 1935/12288 \approx 0.15747$, and $f_3(p) \le 63/400$ for every p, so the loss is below $3 \cdot 10^{-5}$.

Lemma 5. *For every $a \ge 1/2$, $f_{\lceil a \rceil}(1/2) \ge 1/(8a)$.*

Proof. Let $k = \lceil a \rceil \ge 1$. Then $f_k(1/2) = (1 - 2^{-k})/(2k) \ge 1/(4k)$, and $k \le 2a$: if $k = 1$ this follows from $a \ge 1/2$, and if $k \ge 2$ from $k < a + 1 \le 2a$. □

As the density grows, the improvement approaches a factor of 8, provided that p shrinks at the right speed.

Proposition 6. *Let $\kappa_{\mathrm{AS}}(a) = \ln 3 / \ln\big(1/(1 - \frac{1}{8a})\big)$ and $\kappa(a,p) = \ln 3 / \ln\big(1/(1 - f_{\lceil a \rceil}(p))\big)$. If $p = p(a) \in [0,1]$ satisfies $p \to 0$ and $ap \to \infty$ as $a \to \infty$, then $\kappa_{\mathrm{AS}}(a)/\kappa(a,p(a)) \to 8$. This holds for $p = 2^{-j}$ with $j = \lfloor \frac{1}{2} \lfloor \log_2 \lceil a \rceil \rfloor \rfloor$, that is, with j fair coins per part and level. For a fixed $p \in (0,1]$, the ratio tends to $8(1-p)$.*

Proof. Let $k = \lceil a \rceil$, $y = 1/(8a)$ and $f = f_k(p)$. For $0 \le x < 1$ we have $x \le \ln(1/(1-x)) \le x/(1-x)$, so the ratio $\ln(1/(1-f))/\ln(1/(1-y))$ lies between $8af(1-y)$ and $8af/(1-f)$. Here $8af = 8\frac{a}{k}(1-p)\big(1-(1-p)^k\big)$, $a/k \to 1$, and $(1-p)^k \le e^{-pk} \le e^{-ap} \to 0$. Hence $8af \to 8$, and $f = 8af \cdot y \to 0$, so both bounds tend to 8. For a fixed p, $(1-p)^k \to 0$ because $k \to \infty$, and $8af \to 8(1-p)$. For the dyadic choice, $4^j \le k < 4^{j+1}$ gives $p^2 < 4/a$ and $(ap)^2 \ge a^2/(a+1) \ge a/2$ for $a \ge 1$. □

2.1 A sharper count for planar graphs

Theorem 4 charges every edge to one endpoint only. For planar quotients we count every edge at both endpoints. For $\tau > 0$ let Q_τ be the spanning subgraph of the edges of weight at least τ, and let $d_\tau(P)$ be the degree of P in Q_τ.

Lemma 7 (layer cake)**.** *If every vertex of Q is in C independently with probability p, then*

$$\mathbb{E}[\mathrm{gain}(C)] = s \int_0^\infty \sum_P g(d_\tau(P))\, d\tau.$$

Proof. For fixed P, $[P \notin C] \max_{X \in C \cap N(P)} w(PX) = \int_0^\infty [P \notin C]\,[C \cap N_{Q_\tau}(P) \neq \emptyset]\, d\tau$. The event $P \notin C$ has probability s and is independent of $C \cap N(P)$, and $\Pr[C \cap N_{Q_\tau}(P) \neq \emptyset] = g(d_\tau(P))$. Take expectations, exchange them with the integral (the integrand is bounded and vanishes for $\tau > \max w$), and sum over P. □

Lemma 8. *Every simple planar graph with degrees d_v and n' non-isolated vertices satisfies*

$$\textit{(A)}\ \ |E| \le \sum_v \min(d_v, 3), \qquad \textit{(B)}\ \ 2|E| \le \sum_v \min(d_v, 6) + 2n'.$$

Proof. (A) Let $H = \{v : d_v \ge 3\}$. At most $3|H|$ edges lie inside H, because every simple planar graph on m vertices has at most $3m$ edges. Every other edge has an endpoint $v \notin H$, and there are at most $\sum_{v \notin H} d_v = \sum_{v \notin H} \min(d_v, 3)$ of them. (B) Let $H = \{v : d_v \ge 6\}$. Then $\sum_{v \in H} d_v = 2|E(H)| + |E(H, \bar{H})| \le 6|H| + |E(H, \bar{H})|$, so $2|E| = \sum_v \min(d_v, 6) + \sum_{v \in H}(d_v - 6) \le \sum_v \min(d_v, 6) + |E(H, \bar{H})|$. The edges between H and $\bar{H}$ form a simple bipartite planar graph on non-isolated vertices, which has at most $2n'$ edges. □

Lemma 9 (degree inequality)**.** *Let $p \in [21/64, 1]$ and $c(s) = \big(2g(3) + g(6)\big)/9 = (3 - 2s^3 - s^6)/9$. Every simple planar graph satisfies $\sum_v g(d_v) \ge c(s)\,|E|$.*

Proof. Put $\beta = (g(6) - g(3))/3$ and $\alpha = (8g(3) - 5g(6))/9$, so that $\alpha + 2\beta = c(s)$. Here $\beta \ge 0$ because g is increasing, and $\alpha \ge 0$ because $9\alpha = (1 - s^3)(3 - 5s^3)$ and $s^3 \le 3/5$. We claim that for every integer $d \ge 0$,

$$g(d) \ \ge\ \alpha \min(d, 3) + \beta\big(\min(d, 6) + 2[d \ge 1]\big).$$

Equality holds at $d \in \{0, 3, 6\}$. For $d > 6$ the right side is constant and g is increasing. For $d \in \{4, 5\}$ the right side is the chord of the concave function g between 3 and 6. For $d \in \{1, 2\}$ the claim reads $9g(1) \ge 4g(6) - g(3)$ and $9g(2) \ge 4g(3) + 2g(6)$, that is, $h_1(s) = 6 - 9s - s^3 + 4s^6 \ge 0$ and $h_2(s) = 3 - 9s^2 + 4s^3 + 2s^6 \ge 0$. Both polynomials are decreasing on $[0, 43/64]$ ($h_1' \le -9 + 24s^5 < 0$, and $h_2' = 6s(2s + 2s^4 - 3) < 0$), and $h_1(43/64) > 0.017$, $h_2(43/64) > 0.33$. Summing the claim over all vertices and applying Lemma 8 gives $\sum_v g(d_v) \ge \alpha|E| + 2\beta|E|$. □

Theorem 10. *Let Q be a simple planar graph. If every vertex of Q is in C independently with probability $p \in [21/64, 1]$, then*

$$\mathbb{E}[\mathrm{gain}(C)] \ \ge\ \varphi(p)\, w(Q), \qquad \varphi(p) = \frac{(1-p)\big(3 - 2(1-p)^3 - (1-p)^6\big)}{9}.$$

Proof. Every Q_τ is a simple planar graph, and $\int_0^\infty |E(Q_\tau)|\, d\tau = w(Q)$. Apply Lemma 9 to each Q_τ inside Lemma 7. □

The values are $\varphi(3/8) = 357115/2097152 \approx 0.17029$, $\varphi(1/2) = 175/1152 \approx 0.15191$ and $\varphi(21/64) \approx 0.17181$, compared with $f_3(3/8) \approx 0.15747$ and $f_3(1/2) = 7/48$. Over $p \in [0,1]$, φ is maximized at $s^3 = (\sqrt{37}-4)/7$, that is, $p \approx 0.3324$, where $\varphi \approx 0.17182$. The count is tight for its inputs. Over degree sequences that satisfy only (A) and (B), the infimum of $\sum_v g(d_v)/|E|$ is $c(s)$. It is approached by two vertices of degree 3 per vertex of degree 6, together with a vanishing number of hubs incident to two thirds of the edges. The certificate above is the dual solution of this linear program.

2.2 Other classes

Planarity enters Theorem 10 only through Lemma 8, and its proof uses two density bounds. Let $\mathcal{C}$ be a class of graphs that is closed under taking subgraphs, and let $a, b > 0$ be such that every $H \in \mathcal{C}$ has $|E(H)| \le a|V(H)|$ and every bipartite $H \in \mathcal{C}$ has $|E(H)| \le b|V(H)|$. Planar graphs have $a = 3$ and $b = 2$.

Lemma 11. *Every $H \in \mathcal{C}$ with degrees d_v and n' non-isolated vertices satisfies, for all reals $t \ge a$ and $u \ge 2a$,*

$$(A_t)\ \ |E| \le \sum_v \min(d_v, t), \qquad (B_u)\ \ 2|E| \le \sum_v \min(d_v, u) + b\,n'.$$

Proof. As for Lemma 8. At most $a|H_t| \le \sum_{v \in H_t} \min(d_v, t)$ edges lie inside $H_t = \{v : d_v \ge t\}$. For $H_u = \{v : d_v \ge u\}$ we have $\sum_{v \in H_u} d_v \le 2a|H_u| + |E(H_u, \bar{H}_u)| \le u|H_u| + |E(H_u, \bar{H}_u)|$, and the edges between H_u and $\bar{H}_u$ form a bipartite graph in $\mathcal{C}$ on at most n' non-isolated vertices. □

Theorem 12. *Let $s = 1 - p$. Let $\alpha_1, \dots, \alpha_m, \beta_1, \dots, \beta_\ell \ge 0$ and $t_i \ge a$, $u_j \ge 2a$ satisfy, for every integer $d \ge 0$,*

$$g(d) \ \ge \ \sum_i \alpha_i \min(d, t_i) + \sum_j \beta_j \big(\min(d, u_j) + b\,[d \ge 1] \big).$$

If every vertex of a simple graph $Q \in \mathcal{C}$ is in C independently with probability p, then $\mathbb{E}[\mathrm{gain}(C)] \ge s\,c\,w(Q)$ with $c = \sum_i \alpha_i + 2\sum_j \beta_j$.

Proof. Every Q_τ lies in $\mathcal{C}$. Summing the certificate over its vertices and applying Lemma 11 gives $\sum_P g(d_\tau(P)) \ge c\,|E(Q_\tau)|$. Integrate as in Theorem 10. □

Lemma 9 is the case $t = 3$, $u = 6$, $b = 2$. Without any bipartite bound, a single threshold already improves Theorem 4 when a is not an integer.

Corollary 13. *Let $a \ge 1$ and $c_a(s) = \min\{ g(\lfloor a \rfloor)/\lfloor a \rfloor,\ g(\lceil a \rceil)/a \}$. Then $\mathbb{E}[\mathrm{gain}(C)] \ge s\,c_a(s)\,w(Q)$ for every simple $Q \in \mathcal{C}$. For integer a this is $f_a(p)$. For non-integer a and $0 < s < 1$ it is larger than $f_{\lceil a \rceil}(p)$.*

Proof. Take $t = a$ and $\alpha = c_a(s)$ in Theorem 12. For $d \le \lfloor a \rfloor$ we have $\alpha d \le g(d)$, because $g(d)/d$ is non-increasing. For $d \ge \lceil a \rceil$ we have $\alpha a \le g(\lceil a \rceil) \le g(d)$. For non-integer a, both terms of the minimum exceed $g(\lceil a \rceil)/\lceil a \rceil$, the first by strict concavity of g. □

For integer densities, no improvement over Theorem 4 is possible without excluding more structure.

Proposition 14. *Let $k \geq 1$. For every p, on $K_{k,n}$ with unit weights, the expected fraction of the edges that become internal is at most $f_k(p) + 1/n$. Hence, on every class that contains $K_{k,n}$ for all n, no analysis of the random-centre rule gives a per-level fraction above $\max_p f_k(p)$. Such classes include the graphs of treewidth at most k and, for $k \leq 5$, the K_{k+2}-minor-free graphs, both of density k [19].*

Proof. The leaves of a star in a bipartite graph lie on the side opposite to its centre and are pairwise non-adjacent, so the internal edges are exactly the join edges. A vertex of degree k has a join edge exactly when it is not a centre and one of the k hubs is, which has probability $s(1 - s^k)$. The hubs have at most k join edges, and $|E| = kn$. Finally, $K_{k,n}$ has treewidth k, and K_{k+2} has treewidth $k + 1$. □

By Proposition 14, Theorem 4 is optimal for every integer row of Table 1 whose class contains all $K_{a,n}$. In particular, K_5-minor-free graphs have the same density as planar graphs, but the rule cannot beat the exponent 6.41 on them. The improvement to 5.83 for planar graphs comes from excluding $K_{3,n}$, which is what the bipartite bound $b = 2$ records.

Surfaces. A simple graph with $n \geq 3$ vertices that embeds in a surface of Euler genus e has at most $3n - 6 + 3e$ edges, and at most $2n - 4 + 2e$ if it is bipartite. For $e \leq 2$ (the sphere, projective plane, torus and Klein bottle) this gives $a = 3$ and $b = 2$. Hence Theorem 10 and all its numbers, including the exponent 5.89 for $p = 3/8$, hold for graphs on these surfaces. For larger e we use $a_e = \max\{3, \max_n \min\{(n-1)/2,\ 3 - (6 - 3e)/n\}\}$ and $b_e = \max\{2, \max_n \min\{\lfloor n^2/4 \rfloor / n,\ 2 - (4 - 2e)/n\}\}$. Table 2 lists the resulting exponents. Every entry of the last column uses a certificate for Theorem 12 that was found by linear programming (`sim/tight/surfaces.py`). The certificates and the exponent bounds are checked in Lean (Section 6).

Euler genus e	a_e	b_e	[1]	Thm. 4	Cor. 13	Thm. 12
$0, 1, 2$	3	2	25.81	6.42	6.42	5.83
3	27/8	20/9	29.11	7.66	6.42	6.13
4	11/3	12/5	31.67	7.66	6.97	6.55
6	21/5	30/11	36.36	8.88	7.66	7.19
8	51/11	3	40.20	8.88	8.19	7.75
12	11/2	24/7	47.79	10.08	9.19	8.76
20	33/5	70/17	57.46	11.28	10.60	10.10
100	27/2	8	118.10	19.45	18.73	18.17
200	37/2	11	162.04	25.18	24.50	23.93

Table 2: Round exponents $\ln 3 / \ln(1/q)$ for graphs on a surface of Euler genus e (for an orientable surface, e is twice the genus), at the best p with at most 8 fair coins. Our exponents are rounded up.

3 Plugging the rule into the clustering algorithm

We refer to the numbering of [1] and recall its notation. The clustering acts on a weighted graph H of the class $\mathcal{C}$ and maintains partitions $\mathcal{P}_0, \mathcal{P}_1, \ldots$ of $V(H)$ into connected parts, each with a leader. For a partition $\mathcal{P}$, $\mathrm{cut}_w(\mathcal{P})$ is the total weight of the edges of H between different parts, and $H/\mathcal{P}$ is the quotient whose vertices are the parts and whose edge weights add up the weights between

two parts. The σ-algebra $\mathcal{F}_{t-1}$ is generated by the coins of the levels before t, and $T_{\mathcal{C}}$ is as in [1, Thm. 5.4]. In Algorithm 1 (RandomCluster), we replace lines 6–13 at level t as follows. Every part $P \in \mathcal{P}_{t-1}$ reads b level-t coins of its leader and becomes a centre with probability $p = m/2^b$. Every non-centre part with a centre neighbour in the quotient $Q = H/\mathcal{P}_{t-1}$ joins its heaviest centre neighbour, with ties broken by identifier. Every centre absorbs the parts that joined it, and every other part is unchanged.

- *Lemma 4.1* (partition, radius $r_t = 3r_{t-1} + 1$, quotient in $\mathcal{C}$, dependence on coins). The proof only uses that the new parts are vertex-disjoint stars whose leaves are adjacent to their centre in Q, which holds by Lemma 2.

- *Lemmas 4.2 and 4.3.* Fix the coins of levels $< t$. Then Q is fixed and lies in $\mathcal{C}$, so it is a-sparse and has an orientation with out-degree at most $\lceil a \rceil$. The centre events are independent, because distinct parts have distinct leaders. By Lemma 2 every join edge becomes internal, so $\mathrm{cut}_w(\mathcal{P}_t) \le w(Q) - \mathrm{gain}(C)$. Theorem 4 gives $\mathbb{E}[\mathrm{cut}_w(\mathcal{P}_t) \mid \mathcal{F}_{t-1}] \le (1 - f_{\lceil a \rceil}(p))\,\mathrm{cut}_w(\mathcal{P}_{t-1})$. This replaces Lemma 4.3 with $q' = 1 - f_{\lceil a \rceil}(p)$, and Corollary 4.4 follows verbatim. For planar graphs, Q is a minor of a planar graph and has no parallel edges, so Theorem 10 gives $q' = 1 - \varphi(p)$ instead, and for the classes of Section 2.2 Theorem 12 gives $q' = 1 - sc$. Lemma 4.2 is no longer needed.

- *Lemma 4.5* (view radius $\rho_t = 2(3^t - 1)$). Let $P = \mathcal{P}_{t-1}(y)$. The new part of y is determined by the centre status of P, its quotient neighbours C and their quotient neighbours $R \subseteq B_H(y, 6r + 2)$, together with the join choice $c(R)$ of each such R. The choice $c(R)$ depends on the quotient edges at R and on the coins of the parts meeting $B_H(y, 6r + 3)$. The proof in [1] shows that the ρ_t-view of y reveals all of these parts and their level-t coins. Hence ρ_t is unchanged.

- *Theorem 5.4.* The bound $T_{\mathcal{C}} + 9 \cdot 3^L$ on the number of rounds is unchanged. The algorithm uses bL instead of L fair coins per vertex, and the factor q becomes q' in both expectation bounds.

- *Theorem 6.1* (the derandomization of [13]) requires only a prescribed finite number of fair coins per vertex, which still holds.

- *Theorem 6.2 and Corollary 8.1* hold with q' in place of q. For planar graphs, $p = 3/8$ gives $q' = 1740037/2097152$, $p = 1/2$ gives $q' = 977/1152$, and $p = 21/64$ gives $q' = 3642429005643/2^{42}$. This proves Theorem 1.

The same substitution improves the randomized $(1 + \varepsilon)$-approximation and the deterministic fractional $(1 + \varepsilon)$-approximation of [1, Thm. 6.2] for every proper minor-closed class.

4 Limits of the approach

Theorem 10 bounds only the join edges, and this cannot be pushed much further. For unit weights, Lemma 7 gives the exact value $\mathbb{E}[\mathrm{gain}]/w = s \sum_v g(d_v)/|E|$. On the Apollonian network obtained from K_4 by inserting a vertex into every face ten times (118100 vertices, script `apo.py`), this value is at most 0.17402 for every p (the maximum is at $p \approx 0.320$), and it equals 0.17161 at $p = 3/8$, against $\varphi(3/8) = 0.17029$. Deeper networks lower these values slightly. Hence no lower bound on the expected gain of the rule, for any p, exceeds 0.17402, and the exponent of any analysis that counts only join edges is at least 5.74. Theorem 10 gives 5.83.

Counting the edges between leaves of the same star helps only on graphs with triangles. Proposition 14 with $k = 2$ bounds the rule itself, with every internal edge counted: planar graphs contain all $K_{2,n}$, so no analysis of the random-centre rule yields a per-level fraction above $\max_p f_2(p) = 1/(3\sqrt{3}) \approx 0.1925$ on planar graphs, that is, an exponent below 5.13.

No star contraction of the square grid contracts more than a $2/5 + o(1)$ fraction of its edges. The grid is triangle-free, so a star with ℓ leaves internalizes exactly ℓ edges. Each centre has at most 4 leaves, so at most $4n/5$ edges become internal. Any analysis that bounds every level by a uniform worst-case fraction, with radius growth $r \mapsto 3r + 1$, therefore cannot give an exponent below $\ln 3/\ln(5/3) \approx 2.15$.

The two remaining propositions bound every analysis that counts only join edges, as ours does.

Proposition 15. *For every even $k \geq 2$ there is a triangulation G with $n = 4k$ vertices and $\gamma(G) = n/4$. In every partition of G into stars, at most $3n/4$ edges are join edges. Hence, for any one-level star contraction, an analysis that counts only join edges gives at most a $3n/(4(3n-6)) \to 1/4$ fraction per level, and an exponent of at least $\ln 3/\ln(4/3) \approx 3.819$.*

Proof. Let H consist of k nested triangles $a_ib_ic_i$, $0 \leq i < k$, where consecutive triangles are joined by the six edges a_ib_{i+1}, a_ic_{i+1}, b_ia_{i+1}, b_ic_{i+1}, c_ia_{i+1}, c_ib_{i+1}. For even $i < k$, the faces $a_ib_ic_{i+1}$ and $a_{i+1}b_{i+1}c_i$ lie in the band between the triangles i and $i+1$. These k faces are pairwise disjoint and cover $V(H)$. Insert a new vertex into each of them. The result G is a triangulation with $3k+k = n$ vertices. The closed neighbourhoods of the inserted vertices partition $V(G)$, so $\gamma(G) \geq k$, and the inserted vertices form a dominating set. In a partition into stars, the centres and the singleton parts dominate G, so there are at least $\gamma(G)$ parts. Every vertex that is not a centre or a singleton has exactly one join edge, so there are at most $n - \gamma(G) = 3n/4$ join edges, while $|E(G)| = 3n - 6$. □

It was conjectured in [20] that every sufficiently large triangulation satisfies $\gamma \leq n/4$. If the conjecture holds, then Proposition 15 is the best such bound for triangulations.

Proposition 16. *Consider any rule in which every part becomes a centre independently with a probability that depends only on its weighted radius-r neighbourhood, without identifiers, and every non-centre joins a centre neighbour. On the $m \times m$ triangular grid with unit weights, the expected number of join edges is at most $m^2 \max_s s(1-s^6) + O(rm)$. Hence an analysis of such a rule that counts only join edges gives at most $2 \cdot 7^{-7/6} + o(1) \approx 0.2066$ per level, and an exponent of at least 4.74.*

Proof. All vertices at distance more than $r+1$ from the boundary have the same probability p, and so do their six neighbours, because their neighbourhoods are isomorphic. Such a vertex has a join edge with probability $s(1-s^6)$, where $s = 1-p$. Every vertex has at most one join edge, at most $O(rm)$ vertices lie near the boundary, and the grid has $3m^2 - O(m)$ edges. The maximum of $s(1-s^6)/3$ is attained at $s = 7^{-1/6}$. □

5 The dependence on ε

The deterministic $(1+\varepsilon)$-approximation of γ_f and the randomized algorithm with expected size $(1+\varepsilon)\gamma$ run in $(c_0/\varepsilon)^\kappa$ rounds, both in [1] and with the rule of Section 2. Open Problem 3 of [1] asks for the optimal dependence on ε. For integral solutions on general graphs, a $(1+\varepsilon)$-approximation of γ needs $\Omega(\log n/\varepsilon)$ rounds [4]. On every proper minor-closed class, the randomized algorithm of [1] with expected size $(1+\varepsilon)\gamma$ needs a number of rounds that depends only on ε.

5.1 A lower bound on paths

Paths give a lower bound that is linear in $1/\varepsilon$. We let a vertex learn in r rounds its radius-r ball together with all identifiers and degrees in it. This only strengthens the algorithm.

Theorem 17. *Let $r \geq 0$ and $L = 2r + 6 + (r \bmod 3)$. Let $\mathcal{A}$ be an r-round LOCAL algorithm that is either deterministic and outputs a fractional dominating set, or randomized and always outputs a dominating set. Then for some identifier assignment on the path P_L, the (expected) size of the output is at least $\gamma(P_L) + 2/3 = (1 + 2/L)\gamma_f(P_L)$. This holds even if $\mathcal{A}$ knows the graph. Consequently, every r-round algorithm of either kind has ratio at least $1 + 1/(r + 4)$ on paths, and a $(1 + \varepsilon)$-approximation needs at least $1/\varepsilon - 4$ rounds.*

Proof. Let $R = r + 1$ and $M = L - 2R = 4 + (r \bmod 3)$, and number the vertices $0, \ldots, L - 1$. Draw the identifiers as a uniformly random injection into the finite identifier space, independently of the coins of $\mathcal{A}$, and let x_v be the expected value of the output at v. Every outcome is a fractional dominating set, so x is one, and $\sum_v x_v$ is the average of the (expected) output size over the identifier assignments. If an isomorphism from the radius-r ball around u to the one around v maps u to v and preserves the degrees in P_L, then $x_u = x_v$. Indeed, the output at a vertex depends only on its ball with the identifiers, the degrees and the coins in it, and the identifiers and the coins on the two balls have the same distribution. Hence $x_v = x_{L-1-v}$ for all v, and all M middle vertices $R \leq v < R + M$ have the same value μ: their balls are paths with $2r + 1$ vertices of degree 2. Since $M \geq 3$, the closed neighbourhood of $R + 1$ consists of middle vertices, so $\mu \geq 1/3$.

Choose $k \in \{0, 1, 2\}$ and $t \geq 0$ with $R + k = 3t + 2$. The closed neighbourhoods of the vertices $0, 3, \ldots, 3t$ are disjoint. Their union $\{0, \ldots, R + k - 1\}$ consists of the left part $\{0, \ldots, R - 1\}$ and k middle vertices. Hence $\sum_{v<R} x_v + k\mu \geq t + 1 = (R + k + 1)/3$. By the symmetry, and since $M - 2k \geq 0$ and $\mu \geq 1/3$,

$$\sum_v x_v = 2\sum_{v<R} x_v + M\mu \;\geq\; \frac{2(R + k + 1)}{3} + (M - 2k)\mu \;\geq\; \frac{2R + M + 2}{3} = \frac{L + 2}{3}.$$

Finally $3 \mid L$. The vertices $1, 4, \ldots, L-2$ dominate P_L, and their closed neighbourhoods are disjoint, so $\gamma(P_L) = \gamma_f(P_L) = L/3$. The ratio bound follows from $L \leq 2r + 8$. □

The bound holds whether or not the algorithm uses identifiers or knows n. The only obstruction is that the middle vertices cannot see the ends. For fractional domination the bound is close to exact. Every fractional dominating set of P_ℓ has weight at least $\lceil \ell/3 \rceil$. On P_ℓ with $\ell \geq 2$, put $x_v = 2/3$ at the two ends and $x_v = 1/3$ at the other vertices. This is a fractional dominating set of weight $(\ell + 2)/3 \leq (1 + 2/\ell)\lceil \ell/3 \rceil$, and every vertex knows its value from its own degree. If $\ell \leq 2r + 2$, every vertex sees the nearer end within distance r. If it also knows ℓ, it can output the average of a fixed minimum dominating set and its mirror image, which has weight $\lceil \ell/3 \rceil$. With the weights $2/3$ and $1/3$ for $\ell \geq 2r + 3$, an algorithm that knows the length has ratio at most $1 + 2/(2r + 3)$ for every $r \geq 0$. So for fractional algorithms that know the length, the smallest number of rounds for ratio $1 + \varepsilon$, with $0 < \varepsilon \leq 1$, lies between $1/\varepsilon - 4$ and $1/\varepsilon - 1/2$. Without the length, let a vertex output the same average if it sees both ends of a path with at most $r + 1$ vertices, and the weights $2/3$ and $1/3$ otherwise. On a path with at most $r + 1$ vertices every vertex sees both ends, and on a longer path no vertex uses the average. So the output is a fractional dominating set, its ratio is at most $1 + 2/(r + 2)$, and $\lceil 2/\varepsilon \rceil - 2$ rounds suffice for ratio $1 + \varepsilon$. For $r \leq 15$ we computed the optimal ratios numerically. They are $1 + 2/L$ for fractional algorithms that know the length. For those that do not, with identifiers drawn independently of the length, they are $1 + 1/(r + 2)$ if $r \not\equiv 2$ and $1 + 1/(r + 3/2)$ if $r \equiv 2 \pmod 3$ (script `path_sym.py`).

For randomized algorithms that always output a dominating set, the bound is far from exact when r is small. Let $r = 0$, and suppose that three identifiers can each be absent from the output, with positive probability, at a vertex of degree 2. On three consecutive inner vertices of a path with these identifiers, all three are absent with positive probability, and the middle one is not dominated. So at most two identifiers can be absent. If the identifier range has at least $\ell+2$ elements, the inner vertices of P_ℓ can avoid them, so the output has size at least $\ell - 2$, and the ratio tends to 3 as ℓ grows. For these algorithms, the best upper bound on paths we know is the $O(1/\varepsilon)$ of Theorem 19.

Theorem 17 does not cover deterministic algorithms that output dominating sets. In a constant number of rounds, they cannot beat ratio 3 on paths [1, Thm. 8.7].

Corollary 18. *Let $\mathcal{C}$ be a minor-closed class. If $\mathcal{C}$ excludes some path, a minimum dominating set can be computed deterministically in a constant number of rounds. Otherwise, a deterministic $(1+\varepsilon)$-approximation of γ_f on $\mathcal{C}$, or a randomized algorithm that always outputs a dominating set of expected size at most $(1+\varepsilon)\gamma$, needs at least $1/\varepsilon - 4$ rounds. For planar graphs, the optimal number of rounds lies between $1/\varepsilon - 4$ and $O((c_0/\varepsilon)^{5.83})$.*

Proof. If $P_k \notin \mathcal{C}$, every path in a graph of $\mathcal{C}$ has fewer than k vertices, so every component has diameter less than k. In k rounds every vertex sees its component, including all identifiers, and all vertices of a component agree on its lexicographically first minimum dominating set. Otherwise $\mathcal{C}$ contains all paths, since it is closed under subgraphs, and Theorem 17 applies. The upper bound for planar graphs is the six-coin exponent of Theorem 1, applied to the two guarantees of [1, Thm. 6.2] named above (Section 3). □

5.2 Forests: the lower bound is tight

For forests, which have density $a = 1$, Table 1 gives $(c_0/\varepsilon)^{3.82}$ rounds. We show that $O(1/\varepsilon)$ rounds suffice, even with unbounded degrees.

Theorem 19. *There is a constant C such that for every $\varepsilon \in (0, 1]$ the following hold. A deterministic LOCAL algorithm computes, on every forest G, a fractional dominating set of weight at most $(1+\varepsilon)\gamma_f(G)$ in C/ε rounds. A randomized LOCAL algorithm always outputs a dominating set of expected size at most $(1+\varepsilon)\gamma(G)$ in C/ε rounds.*

With Theorem 17, the optimal number of rounds on forests is $\Theta(1/\varepsilon)$, both for the deterministic fractional and for the randomized $(1+\varepsilon)$-approximation.

Proof. Repair. For $U \subseteq V$ let $\mathrm{opt}(U)$ be the smallest weight of an $x \geq 0$ supported on U with $x(N[u]) \geq 1$ for all $u \in U$. Let $B \subseteq U$ meet $N[u]$ for every $u \in U$ with a neighbour outside U. For every fractional dominating set y of G,

$$\mathrm{opt}(U) \leq y(U) + |B|. \tag{1}$$

Indeed, y on U plus weight 1 on B is feasible. The integral version of (1), for dominating sets, holds in the same way.

Core and hairs. Let F be a forest and $R \geq 1$. For an edge vw of F, let $h(v, w)$ be the largest distance from v to a vertex of the component of $F - v$ that contains w. The *core* K consists of the vertices v with $h(v, w) \geq R$ for at least two neighbours w.

(a) *K meets every component of F in a connected set.* Let $v, v' \in K$ and let z lie on the path between them. One of the two high branches of v avoids z, so the branch of z towards v has height at least R, and likewise towards v'.

(b) *Every vertex outside K in a component that meets K has distance at most R to K.* Let $u_0, \dots, u_k$ be a shortest path from $u_0 \notin K$ to $u_k \in K$. The branch of u_i towards u_0 has height at least i. The branch towards u_k has height at least R, because it contains a high branch of u_k that avoids u_{k-1}. So $u_i \in K$ for $R \le i \le k$, and minimality gives $k \le R$. Let $a(u)$ be the nearest core vertex, and let $\mathrm{hair}(c) = \{c\} \cup a^{-1}(c)$; by (a), $a(u)$ is unique. Every neighbour w of a vertex $u \in \mathrm{hair}(c) \setminus \{c\}$ lies in $\mathrm{hair}(c)$: if $w \in K$ then $a(u) = w$, and otherwise the path from one of u, w to K passes through the other.

(c) *Components of F that miss K have diameter less than $2R$.* On a path of length at least $2R$, the vertex at distance R from one end would lie in K.

(d) *A core vertex v with at most one core neighbour has a path $v, x_1, \dots, x_R$ with $x_1, \dots, x_R \in \mathrm{hair}(v)$.* One of the high branches of v starts at a vertex outside K. By (a), this branch contains no core vertex, so it lies in $\mathrm{hair}(v)$, and it contains a vertex at distance R from v.

Call a core vertex a *branch vertex* if it has no core neighbour or at least three. The other core vertices induce paths, the *core paths*. Let L be the number of core vertices with at most one core neighbour, and let d_K be the degree in $F[K]$. Every core vertex v satisfies

$$[v \text{ branch}] + 2 \le d_K(v) + 3\,[d_K(v) \le 1], \qquad [v \text{ branch}]\, d_K(v) + 6 \le 3d_K(v) + 6\,[d_K(v) \le 1],$$

as one checks for $d_K(v) = 0$, 1, 2 and $d_K(v) \ge 3$. Since $F[K]$ is a forest, $\sum_{v \in K} d_K(v) \le 2|K|$. Summing over K, there are at most $3L$ branch vertices, and $\sum_b d_K(b) \le 6L$ over the branch vertices b.

Deterministic version. Apply this to $F = G$, and let P be the set of vertices of the core paths. For $u \in P$ and $0 \le k \le R$, let $P(u,k)$ be the set of vertices of the core path of u at distance at most k from u, and let $s(u) \in \{0,1,2\}$ be 2 minus the number of neighbours of u on its core path. For $X \subseteq K$ let $\mathrm{hair}(X) = \bigcup_{c \in X} \mathrm{hair}(c)$. The windows are:

- $\mathrm{hair}(P(u,R))$ for every $u \in P$, with weight $1/(2R+1)$;
- $\mathrm{hair}(P(e, R-t))$ for every $e \in P$ and $1 \le t \le R$, with weight $s(e)/(2R+1)$;
- $\mathrm{hair}(b)$ for every branch vertex b, with weight 1;
- every component of G that misses K, with weight 1.

On a core path $p_1 \dots p_m$, the windows of the first two kinds are the hairs of the sets $\{p_j : |j-i| \le R\}$ for $1 - R \le i \le m + R$, each with weight $1/(2R+1)$. Those with $i \le 0$ are the windows at $e = p_1$ with $t = 1 - i$, and those with $i > m$ are the windows at $e = p_m$ with $t = i - m$; if $m = 1$, then $s(p_1) = 2$ accounts for both. So every vertex of P lies in windows of total weight exactly 1, and by (b) so does every vertex of its hair. A vertex of the hair of a branch vertex lies only in that window, and a vertex of a component that misses K only in that component. So every vertex lies in windows of total weight exactly 1.

Let ∂U be the set of vertices of U with a neighbour outside U. By (b), the neighbours of a vertex of $\mathrm{hair}(c) \setminus \{c\}$ lie in $\mathrm{hair}(c)$, and so does every neighbour of c outside K. Hence a vertex of $\partial\,\mathrm{hair}(X)$ lies in X and has a core neighbour outside X. For $X = P(u,k)$, it is one of the at most two vertices at distance exactly k from u, or it lies in the set A of vertices of P with a branch neighbour. For $X = \{b\}$ it is b, and the components have no boundary. Each vertex of A lies

in windows of the first two kinds of total weight 1, and all these windows together have weight $(|P| + RS)/(2R+1)$, where $S = \sum_{e \in P} s(e)$. So

$$\sum_U \mathrm{wt}(U)\,|\partial U| \le |A| + \frac{2(|P| + RS)}{2R+1} + 3L \le |A| + \frac{|P|}{R} + S + 3L.$$

Every vertex of A is an end of a core edge at a branch vertex, so $|A| \le 6L$. The number of neighbours of $e \in P$ on its core path is $d_K(e)$ minus its number of branch neighbours. So $s(e)$ is $[d_K(e) = 1]$ plus the number of branch neighbours of e, and $S \le L + 6L = 7L$.

Now let y be a fractional dominating set. For every core vertex c with at most one core neighbour, take the path $c, x_1, \dots, x_R$ of (d). By (b), $N[x_j] \subseteq \mathrm{hair}(c)$. A vertex w lies in $N[x_j]$ only for the c with $w \in \mathrm{hair}(c)$, and for at most three values of j, since the x_j with $w \in N[x_j]$ are at distance at most 2 from each other. So $RL \le \sum_{c,j} y(N[x_j]) \le 3y(V)$. Let I be the set of vertices of P with two core neighbours and no branch neighbour. A vertex $w \notin K$ lies in $N[z]$ with $z \in I$ only for $z = a(w)$, a branch vertex lies in no such $N[z]$, and a vertex of P lies in $N[z]$ only for itself and its at most two neighbours on its core path. So $|I| \le \sum_{z \in I} y(N[z]) \le 3y(V)$, and $|P| \le |I| + L + |A| \le 3y(V) + 7L$. Altogether, since $L \le 3y(V)/R$ and $R \ge 1$,

$$\sum_U \mathrm{wt}(U)\,|\partial U| \le 16L + \frac{3y(V) + 7L}{R} \le \frac{48 + 3 + 21}{R}\, y(V) = \frac{72}{R}\, y(V).$$

Let x^U be the optimal solution of the problem defining $\mathrm{opt}(U)$ that minimizes $\sum_v x_v^2$; it is unique, so the vertices of U agree on it. Output $x = \sum_U \mathrm{wt}(U) x^U$. A vertex u is dominated in each window that contains it, and these windows have total weight 1. By (1) with $B = \partial U$, for every fractional dominating set y,

$$\sum_v x_v \le \sum_U \mathrm{wt}(U)\big(y(U) + |\partial U|\big) \le \Big(1 + \frac{72}{R}\Big) y(V).$$

Take y optimal and $R = \lceil 72/\varepsilon \rceil$. A vertex decides membership in K within radius R, membership in P within $R+1$, the value $s(\cdot)$ within $R+2$, and its nearest core vertex within $2R$. A window that contains v has its centre within $2R$ of v and all its vertices within $4R$ of v, by (b) and (c). So v finds the windows that contain it, and their problems, in its ball of radius $7R$, and the algorithm runs in $7R \le 511/\varepsilon$ rounds.

Randomized version. In $O(1)$ rounds, compute a dominating set D_0 with $|D_0| \le c\gamma(G)$ for an absolute constant c [16]. Every vertex $u \notin D_0$ picks a neighbour $d(u) \in D_0$, and $d(v) = v$ for $v \in D_0$. Contracting the stars $S_d = d^{-1}(d)$ gives a forest H on D_0, and a round on H takes $O(1)$ rounds on G. Let $N = |D_0|$. For $W \subseteq V(H)$ let $V_W = \bigcup_{d \in W} S_d$, and let ∂W be the set of $d \in W$ with an H-neighbour outside W. If $u \in V_W$ has a neighbour $w \notin V_W$, then $d(u)d(w)$ is an H-edge leaving W, so $d(u) \in \partial W \cap N[u]$. So (1) applies to $U = V_W$ and $B = \partial W$.

Take the core and the hairs of $F = H$. By (d), the hair of a core vertex with at most one core neighbour has more than R vertices. The hairs are disjoint, so $L \le N/R$. Cut the core edges at branch vertices; there are at most $6N/R$ of them. Every vertex of a core path draws an independent uniform value in $[0,1]$, given by the binary expansion of a sequence of its coins. Ties have probability 0, and we break them by the identifiers. For a vertex u of a core path, let $B(u)$ be the set of vertices of its core path at distance at most R from u, and let the *anchor* of u be the vertex of $B(u)$ with the smallest value. Cut every edge uv of a core path whose ends have different anchors. This is a one-dimensional low-diameter decomposition in the spirit of [21]. The clusters are the resulting components of $H[K]$ with their hairs, together with the components

that miss K. All core vertices of a cluster have the same anchor, which has distance at most R from each of them. By (b), every vertex of a hair has distance at most R from its core vertex, and by (c), the components that miss K have diameter less than $2R$. So every cluster lies within distance $2R$ in H of one vertex: the common anchor of its core vertices, its branch vertex, or any of its vertices. If $B(u) \neq B(v)$, then $B(u) \cup B(v)$ has at least $R+2$ vertices, and $B(u) \triangle B(v)$ has at most two. The anchors differ only if the smallest value in $B(u) \cup B(v)$ lies in $B(u) \triangle B(v)$, so uv is cut with probability at most $2/(R+2)$. The core paths have fewer than N edges, so the expected number of cut edges is at most $8N/R$. Each cluster W outputs the lexicographically first smallest set $D_W \subseteq V_W$ that dominates every vertex of V_W. Every cut edge puts at most two vertices into the sets ∂W. By the integral form of (1), the expected size is at most $\gamma + 16N/R \leq (1+\varepsilon)\gamma$ for $R = \lceil 16c/\varepsilon \rceil$. In H, a vertex decides membership in K and in P within radius $R+1$, and its nearest core vertex within $2R$, as in the deterministic version. So the anchor of a vertex u of a core path is determined by the ball of radius $2R+1$ around u, and whether an edge uw of a core path is cut, by the ball of radius $2R+2$ around u. The cluster of a vertex d of H lies within distance $4R$ of d, so d determines it, and the edges that leave it, from its ball of radius $6R+2$ in H. A round on H takes $O(1)$ rounds on G, so with the computation of D_0, the algorithm runs in $O(R) = O(1/\varepsilon)$ rounds. □

The numerical checks are in `tree_frac.py` and `tree_ldd.py`. The proof uses forests only to build windows of radius $O(R)$ and weighted boundary $O(\gamma_f/R)$ locally. For planar graphs, such a local family of windows, with no dependence on n or on the maximum degree, would give $\Theta(1/\varepsilon)$ for the fractional approximation in the same way, and polylogarithmic losses would give $\tilde{O}(1/\varepsilon)$. The star contractions of Section 2 give radius $(c_0/\varepsilon)^\kappa$.

5.3 Outerplanar graphs

On outerplanar graphs, the inner faces take the place of the vertices of a tree. Short faces are seen locally, long faces are not. We remove the long faces by passing to a covering graph in which the face structure is a tree, and run the construction of Theorem 19 there. Vertices of high degree are handled by the way core paths are measured.

Theorem 20. *There is a constant C such that for every $\varepsilon \in (0,1]$ the following hold on every outerplanar graph G. A deterministic LOCAL algorithm computes a fractional dominating set of weight at most $(1+\varepsilon)\gamma_f(G)$ in C/ε rounds. A randomized LOCAL algorithm always outputs a dominating set of expected size at most $(1+\varepsilon)\gamma(G)$ in C/ε rounds.*

With Theorem 17, the optimal number of rounds for both $(1+\varepsilon)$-approximations on outerplanar graphs is therefore $\Theta(1/\varepsilon)$, with no dependence on the maximum degree. On arbitrary graphs of maximum degree Δ, a deterministic $(1+\varepsilon)$-approximation of γ_f takes $O(\varepsilon^{-4}\log^2\Delta)$ rounds [14]. In a constant number of rounds, the best ratio of a deterministic algorithm that outputs a dominating set of an outerplanar graph is 5 [3].

Proof. Let $n = |V(G)|$ and $m = |E(G)|$, and let y be an optimal fractional dominating set of G. As in the deterministic part of the proof of Theorem 19, we work in G itself, here for both versions. We repeatedly use the *packing bound*: if the closed neighbourhoods of the vertices of a set A are pairwise disjoint, then $\gamma_f(G) \geq |A|$, since each of them has y-weight at least 1.

Faces. Fix an outerplanar embedding. Every vertex lies on the outer face, so the interior of a cycle C contains no vertex. It is cut by the chords of C into inner faces, all of whose vertices lie on C, and these faces form a tree under sharing an edge. A chord of the boundary of an inner face

would enclose a vertex. Hence the inner faces are exactly the induced cycles of G, independently of the embedding. They are also isometric: a part of G outside an inner face F is attached to F at a vertex or at the two ends of an edge, so a path that leaves F returns through the vertex or edge by which it left. Two inner faces share at most one edge, and two faces with a common edge share no further vertex, since the common chord separates their other vertices. Fix $\ell \geq 14$. An inner face is *short* if it has at most ℓ vertices and a *hole* otherwise. Short faces are recognized in ℓ rounds. An *ℓ-block* is a class of short faces under the equivalence relation generated by sharing an edge, together with the vertices and edges of its faces. An edge on no short face is *thin*. The ℓ-blocks and the thin edges are the *pieces*, and every edge lies in exactly one piece. The faces of an ℓ-block B that contain a vertex v form a path in the weak dual of B, since the weak dual of the block of G that contains them is a tree. By the same uniqueness, two short faces at v lie in one ℓ-block if and only if they are joined by a sequence of short faces at v in which consecutive faces share an edge at v. So the pieces at v are determined by the faces at v, which have length at most ℓ or are seen to be longer within distance ℓ. Finally, every cycle C of length at most ℓ lies in one ℓ-block: the faces inside C are short, they are connected through shared edges, and every edge of C lies on one of them.

The decomposition graph. Let D be the graph with the following nodes and bags: every short face, with its vertex set as bag; every thin edge, with its two ends; every vertex v that lies in no piece or in at least two pieces, with bag $\{v\}$. Two faces are adjacent if they share an edge. A vertex node v is adjacent to the thin edges at v and, for every ℓ-block B at v, to one face of B at v, chosen by the identifiers. The bags of two adjacent nodes meet in their *adhesion*, which is one vertex or one edge. Every edge of G lies in a bag. For every vertex v, the nodes whose bags contain v form a finite tree D_v. Let f be the number of inner faces, f_s the number of short faces, t the number of thin edges, and let p_v be the number of pieces at v. An ℓ-block B with n_B vertices, m_B edges and f_B faces satisfies $n_B - m_B + f_B = 1$ and contains $f_B - 1$ adjacencies of D. Hence $\sum_P (|V(P)| - 1) = m - f_s$ over the pieces P. Moreover $\sum_v (p_v - 1) = \sum_P |V(P)| - n$, and vertices with $p_v = 1$ are not nodes. Together with $n - m + f = c(G)$, where c counts components, and $c(D) = c(G)$, this gives

$$|E(D)| - |V(D)| + c(D) = \sum_P (|V(P)| - 1) - n + c(G) = m - f_s - n + c(G) = f - f_s.$$

So the cycle rank of D equals the number h of holes.

Few holes. We show $h \leq 3\gamma_f/(\ell - 13)$. The inner faces, the bridges and the cut vertices of G form a forest $\mathcal{T}$: two faces are adjacent if they share an edge, and a cut vertex is adjacent to the bridges at it and to one face of every other block at it. For distinct holes F and F' in one component, the path from F to F' in $\mathcal{T}$ leaves F through an edge or a vertex s of F: a chord of a block, with F and F' on different sides, or a cut vertex. So s separates $F - s$ from $F' - s$, and $F \cap F' \subseteq s$. Let $S(F)$ be the set of these separators of F. Let $\mathcal{T}_h$ be the union of the paths in $\mathcal{T}$ between holes. Every separator in $S(F)$ is determined by the node after F on such a path, which is a neighbour of F in $\mathcal{T}_h$. All leaves of the forest $\mathcal{T}_h$ are holes, so the degrees of the holes in $\mathcal{T}_h$ sum to at most $2h$, and in particular $\sum_F |S(F)| \leq 3h$. Let $W(F)$ be the set of vertices of F at distance at least 2 from every element of $S(F)$. Since F is an induced isometric cycle, $W(F)$ has at least $|F| - 4|S(F)|$ vertices, and it contains at least $(|F| - 4|S(F)| - 2)/3$ vertices at pairwise distance at least 3. Let $u \in W(F)$ and $u' \in W(F')$ for distinct holes in one component, and let s be the separator of F towards F'. Then $u' \notin s$, since otherwise $u' \in F \cap F'$ would lie in the separator of F' towards F. So every path from u to u' passes through s, and $d(u, u') \geq 3$. By the packing bound, $\gamma_f \geq \sum_F (|F| - 4|S(F)| - 2)/3 \geq h(\ell - 13)/3$.

Unfolding. The construction below is local, and all bounds in this proof add up over the components of G, so we may assume that G is connected. Then D is connected. Let T be the universal cover of D, a tree, and let Π be its group of deck transformations. It acts freely on the nodes of T, and it is a free group, so no element other than 1 has finite order. Every node of T carries the bag of its image in D. Let $\hat{G}$ be the graph whose vertices are the pairs (s, v) with $s \in V(T)$ and v in the bag of s, where (s, v) and (s', v) are identified whenever ss' is an edge of T and v lies in its adhesion. Its edges join (s, u) and (s, w) whenever uw is an edge of G in the bag of s. Let $\pi(s, v) = v$.

(i) *π is a covering map, and Π acts on $\hat{G}$ freely and transitively on every fibre.* The class of (s, v) consists of the pairs (s', v) with s' in the copy of D_v in T that contains s, because D_v is a tree. Every edge vw of G lies in one piece, and within an ℓ-block in one face or in two adjacent faces of D_v. So the class has exactly one neighbour over every neighbour of v. The group Π permutes the copies of D_v freely and transitively.

(ii) *T is a tree decomposition of $\hat{G}$.* Its adhesions have at most two vertices, each bag has diameter at most $\ell/2$ in $\hat{G}$, and every vertex lies in finitely many bags. Every adhesion separates $\hat{G}$, so a shortest path from an adhesion X to a vertex on one side of X stays on that side. Likewise, a path between two vertices of a bag that leaves the bag returns through the adhesion by which it left. So the bag of a face is an isometric cycle of $\hat{G}$.

(iii) *Two distinct lifts of a vertex have distance more than ℓ.* We claim that every closed walk of length at most ℓ lifts, from every starting point, to a closed walk. We use induction on the length. A backtrack lifts to a backtrack. A closed walk $w_0 \dots w_k$ in which some vertex occurs twice among $w_0, \dots, w_{k-1}$ splits into two shorter closed walks. Otherwise the walk is a cycle, so it lies in an ℓ-block B. Its lift stays in the copy of B that contains the starting point, and this copy maps isomorphically onto B. Now let $\hat{v} \neq \hat{v}'$ lie over v. A path between them of length at most ℓ would project to a closed walk whose lift from $\hat{v}$, by unique path lifting, ends at $\hat{v}'$.

(iv) *Locality.* Let $r \geq 0$, let $\hat{v}$ be a lift of v, and let D_r be the subgraph of D formed by the nodes whose bags meet the ball of radius r around v and the edges whose adhesions meet it. The nodes whose bags contain a vertex u, and the edges of D whose adhesions contain u, are determined by the faces of length at most ℓ at u, and so by the ball of radius ℓ around u. So v computes D_r in $r + \ell$ rounds. The graph D_r is connected, since every D_u is a tree and every edge of G lies in a bag. Let T_r be the component of the preimage of D_r in T that contains the nodes whose bags contain $\hat{v}$. It is a tree that covers D_r, so it is the universal cover of D_r. If $\hat{u}$ has distance at most r from $\hat{v}$, then T_r contains all nodes whose bags contain $\hat{u}$, since they form a copy of D_u and a shortest path from $\hat{v}$ to $\hat{u}$ links it to the copy of D_v. An edge of T between two nodes whose bags meet the ball has an adhesion that meets the ball, since adhesions separate, so it lies in T_r as well. Gluing the bags of T_r as in the definition of $\hat{G}$ therefore gives the ball of radius r around $\hat{v}$, with the nodes of T whose bags meet it and the edges of T between them. Vertices farther away may be split, which does not matter. So v computes all of this in $r + \ell$ rounds.

Core and hairs. All distances are in $\hat{G}$. For an edge ab of T with adhesion X, the *side* of ab at b is the component of $T - ab$ that contains b. Let $h(a, b) \in \mathbb{N} \cup \{\infty\}$ be the supremum of the distances from X to the vertices in its bags. The pair (a, b) and this side are *high* if $h(a, b) \geq R$. By (ii), heights are monotone: if the side of cd at d is contained in the side of ab at b, then $h(a, b) \geq h(c, d)$. Also by (ii), a shortest path from X to a vertex on the side at b stays on that side, so $h(a, b) \geq R$

if and only if some vertex on that side has distance exactly R from X. The core K consists of the nodes a with $h(a,b) \geq R$ for at least two neighbours b. So whether a node lies in K is determined by the ball of radius $R+\ell$ around any vertex of its bag, together with the nodes of T whose bags meet that ball. All of this is invariant under Π, which preserves bags and identifiers.

(a) *K meets every component of T in a connected set.* The proof is that of Theorem 19(a). So every node outside K in a component that meets K has a unique nearest core node $a(\cdot)$. Let $\mathrm{hair}(c) = \{c\} \cup a^{-1}(c)$. As in Theorem 19(b), an edge of T that leaves $\mathrm{hair}(c)$ joins c to a core node.

(b) *Every vertex in a bag of* $\mathrm{hair}(c)$ *has distance less than $R+\ell/2$ from the bag of c.* Let q be the neighbour of c towards a node of $\mathrm{hair}(c)$ whose bag contains the vertex. The side of qc that contains c contains a high side of c, so $h(q,c) \geq R$. Since $q \notin K$, we get $h(q,q') < R$ for all other neighbours q' of q. Each adhesion of q lies in its bag, which has diameter at most $\ell/2$.

(c) *Components of T that miss K are finite, and their vertices span radius less than $R+\ell$.* Along an infinite ray, the adhesions leave every ball by (ii), so far out both directions are high. In a finite component, follow high directions without backtracking. The walk ends at a node with no high direction, or at an edge that is high in both directions and has no other high direction at its ends.

(d) *Hairs are disjoint and inject into G.* If $\mathrm{hair}(c)$ and $\mathrm{hair}(c')$ contain nodes s and gs with $g \in \Pi$, then $gc = c'$. Fix an ownership, that is, a node of D_v for every v, chosen by the identifiers in the bags of D_v, and lift it. Then the vertices owned by the hairs of pairwise inequivalent core nodes project to disjoint sets, and injectively.

(e) *Hairs are local.* Let z be a vertex in the bag of a node $s \notin K$. The component of $T - K$ that contains s lies in $\mathrm{hair}(c)$ for its only core neighbour c, or it is a component of T that misses K. By (b) and (c), all vertices in its bags have distance less than $2R+2\ell$ from z. So the hair that contains s is determined by the ball of radius $3R+3\ell$ around z, together with the nodes of T whose bags meet it.

Runs. A core node is a *branch node* if it has a number of core neighbours other than 2. The other core nodes have exactly two core neighbours, so the components of the subgraph of T that they induce are paths, the *core paths*. For a core path P, let Π_P be the subgroup of the $g \in \Pi$ with $gP = P$. If p and gp lie on P, then $g \in \Pi_P$, since gP is a core path that meets P. Every element of Π_P acts on P as an automorphism. A one-sided infinite path has no automorphism other than the identity. On a finite or two-sided infinite path, every automorphism other than a translation fixes a node or exchanges the ends of an edge, and then its square fixes a node. Since Π acts freely on the nodes and has no elements of finite order, every $g \neq 1$ in Π_P acts on P as a translation. So if P is finite, then $\Pi_P = 1$, and P maps injectively into D. If P is infinite, then it contains nodes p and gp with $g \neq 1$, since D is finite. So P is infinite in both directions, and Π_P is infinite cyclic, generated by a translation by some number $q \geq 1$ of nodes. Two nodes of P have the same image in D if and only if they lie in one orbit of Π_P, and we call q consecutive edges of P a *period*. Number the nodes of a core path as p_i, with i in an interval of $\mathbb{Z}$, and let X_i be the adhesion of the edge $e_i = p_i p_{i+1}$. For a vertex v, the indices i with $v \in X_i$ form an interval, because the nodes whose bags contain v form a subtree. If this interval is nonempty, it is the *run* of v, and its *length* is its number of indices. Every index lies in the runs of the at most two vertices of its adhesion. A run is *maximal* if no other run strictly contains it, and every index lies in a maximal run. Two vertices may have the same run, and we count runs as intervals of indices. Let ρ_i be the length of

the shortest maximal run that contains i. For a node p of a core path, with adhesions X and X' towards its two core neighbours, let $\lambda(p) = d(X, X')$ if $X \cap X' = \emptyset$, and $\lambda(p) = 0$ otherwise. Give e_i the length

$$\mu(e_i) = \frac{1}{\rho_i} + \frac{\lambda(p_i) + \lambda(p_{i+1})}{2},$$

which does not depend on a direction and is invariant under Π. So μ takes finitely many values, all positive. A vertex of high degree may lie in the adhesions of many consecutive edges, but these edges are then short. The bags that contain a vertex x lie within distance $\ell/2$ of x. So by the locality of K, the runs of the vertices of X_i, and with them ρ_i and $\mu(e_i)$, are determined by the ball of radius $R + 2\ell$ around a vertex of X_i, however long these runs are. Let $I_1, I_2, \dots$ be the maximal runs, ordered by their left ends. None contains another, so their right ends are ordered in the same way. No index lies in three of them, so $I_c \cap I_{c+2} = \emptyset$.

(R1) *If $i \in I_c$ and $j \in I_{c'}$ with $c' \geq c + 3$, then $X_i \cap X_j = \emptyset$.* The run of a common vertex would contain the right end of I_c and the left end of I_{c+3}. It would lie in a maximal run that contains the right end of I_c, that is, in I_c or in I_{c+1}. Neither meets I_{c+3}.

(R2) *If k maximal runs meet an interval J of indices, then $(k-4)/2 \leq \sum_{i \in J} 1/\rho_i \leq k$.* For the upper bound, $1/\rho_i \leq \sum_{I \ni i} 1/|I|$ over the maximal runs I. For the lower bound, $\sum_{i \in I} 1/\rho_i \geq 1$ for every maximal run I, since $\rho_i \leq |I|$ for $i \in I$. All but at most four of the k runs lie in J, since a run that sticks out contains an end of J, and every index lies in at most two of them.

(R3) *For $a \leq b$, $d(X_a, X_b) \leq 2\sum_{i=a}^{b} \mu(e_i) + 3$.* Let $I_1, \dots, I_k$ be the maximal runs that meet $[a, b]$, in order, and let v_c be a vertex with run I_c. Then $v_1 \in X_a$ and $v_k \in X_b$, since the first of these runs contains a and the last contains b. If I_c and I_{c+1} share an index, then $d(v_c, v_{c+1}) \leq 1$. Otherwise I_c ends at some i and I_{c+1} starts at $i+1$, where $a \leq i < i+1 \leq b$ since both runs meet $[a, b]$. Among the maximal runs, only I_c contains i, so $X_i \cap X_{i+1} = \emptyset$. Then $d(v_c, v_{c+1}) \leq \lambda(p_{i+1}) + 2$, and $\lambda(p_{i+1}) \geq 1$ is counted in full in $\mu(e_i) + \mu(e_{i+1})$. Hence $d(X_a, X_b) \leq k - 1 + 2\sum \lambda(p_{i+1})$, and $k - 1 \leq 2\sum_{i=a}^{b} 1/\rho_i + 3$ by (R2).

Windows. On every core path, fix ψ with $\psi(p_{i+1}) = \psi(p_i) + \mu(e_i)$. The windows are:

- for every core path and every real τ for which the following set is nonempty, the union of $\mathrm{hair}(p)$ over its nodes with $\tau \leq \psi(p) \leq \tau + R$, with weight $d\tau/R$;
- $\mathrm{hair}(b)$ for every branch node b, with weight 1;
- every component of T that misses K, with weight 1.

Since μ takes finitely many positive values, in every bounded range of τ the first family changes only finitely often, so sums over windows are finite sums of integrals. These families do not depend on the numbering of the core paths, neither on the origin nor on the direction, and they are invariant under Π. Every node of T lies in windows of total weight exactly 1. We identify a window with its node set Q and with the set W of vertices owned by Q. If a path window has the core nodes $p_a, \dots, p_b$, then $\sum_{i=a}^{b-1} \mu(e_i) \leq R$. By (R3), every X_i with $a \leq i < b$ lies within distance $2R + 3$ of X_a. Every bag of $p_{a+1}, \dots, p_b$ contains one of these adhesions and has diameter at most $\ell/2$. With (b), every vertex of the window has distance less than $3R + \ell + 4$ from a fixed vertex of X_a, or of the bag of p_a if $a = b$. By (b) and (c), the other windows have diameter less than $2R + 2\ell$. So every window has diameter less than $6R + 2\ell + 8$.

Output. For a finite $W \subseteq V(\hat{G})$ let $\mathrm{opt}^+(W)$ be the smallest weight of an $x \geq 0$ supported on $N[W]$ with $x(N[u]) \geq 1$ for all $u \in W$, and let x^W be the optimal solution that minimizes $\sum_v x_v^2$.

Then $x^{gW} = g\,x^W$. Vertex v picks a lift $\hat{v}$ and outputs $x_v = \sum_W \mathrm{wt}(W)\,x^W_{\hat{v}}$ over the windows with $\hat{v} \in N[W]$. The value does not depend on the lift. Since π maps $N[\hat{u}]$ bijectively onto $N[u]$,

$$x(N[u]) = \sum_{\hat{w}\in N[\hat{u}]} \sum_W \mathrm{wt}(W)\,x^W_{\hat{w}} \;\geq\; \sum_{W\ni\hat{u}} \mathrm{wt}(W)\,x^W(N[\hat{u}]) \;\geq\; 1.$$

If $gQ = Q$ for a window, then g permutes the finite set Q, so g^k fixes a node for some $k \geq 1$, and hence $g^k = 1$ and $g = 1$. So distinct elements of Π map a window to distinct windows, and the pairs (v, W) with $\hat{v} \in N[W]$ correspond to the pairs of a Π-orbit $[W]$ and a vertex of $N[W]$. Hence $\sum_v x_v = \sum_{[W]} \mathrm{wt}(W)\,\mathrm{opt}^+(W)$.

Repair. Let $\hat{y} = y \circ \pi$. For a window with node set Q, let ∂Q be the set of edges of T with exactly one end in Q. We claim $\mathrm{opt}^+(W) \leq \hat{y}(W) + \sum_{e\in\partial Q} |X_e|$, where X_e is the adhesion of e. Let $z \in W$ have a neighbour $w \notin W$. Walk in T from the owner of z to a node whose bag contains zw through nodes containing z, and then to the owner of w through nodes containing w. The first edge of this walk that leaves Q lies in ∂Q, and its adhesion contains z or w. Both lie in $N[W]$ and dominate z. Pick one of them for every such z. The picked vertices lie in the adhesions of the edges in ∂Q, and weight 1 on them together with $\hat{y}$ on W is feasible. Summing over orbits, and using $\sum_{[W]} \mathrm{wt}(W)\,\hat{y}(W) = \sum_v y_v$, we get

$$\sum_v x_v \leq \gamma_f(G) + 2\sum_{[e]} c(e),$$

where $c(e)$ is the total weight of the windows with $e \in \partial Q$, and $[e]$ ranges over the Π-orbits of edges of T, that is, over $E(D)$.

Counting. By (a), $c(e) = 0$ unless e joins two core nodes. If e lies on a core path, then $c(e) \leq 2\mu(e)/R$. If e meets a branch node, then $c(e) \leq 2$. The core projects to a subgraph K_D of D, and the number of core neighbours of a node equals its degree in K_D. For a finite graph, $\sum_{\deg\geq 3}(\deg -2)$ is at most twice the cycle rank plus the number of vertices of degree 1. Here $\mathrm{rk}(K_D) \leq \mathrm{rk}(D) = h$. So the branch nodes of D have at most $6h + 4N_1$ core neighbours in total, where N_1 is the number of core nodes of D with at most one core neighbour. Such a core node c has a high side whose first node is not in K. By (a), this side lies in $\mathrm{hair}(c)$. It contains a shortest path $u_0, \ldots, u_R$ from its adhesion, and the at least $(R-2)/3$ vertices $u_2, u_5, u_8, \ldots$ have pairwise distance at least 3. Their closed neighbourhoods consist of vertices owned by $\mathrm{hair}(c)$. By (d) and the packing bound, $N_1 \leq 3\gamma_f/(R-2)$. A finite core path ends at two core edges that meet branch nodes. An infinite one projects onto a component of K_D that is a cycle, since each of its nodes has its two core neighbours on it, and infinite core paths in distinct orbits give distinct components. So there are at most $4h + 2N_1$ core paths up to Π.

Next we bound $\sum \mu(e)$ over the edges e of D on core paths. Fix a core path P, and let k_P be its number of maximal runs, up to Π_P if P is infinite. As in the proof of (R2), $1/\rho_i \leq \sum_{I\ni i} 1/|I|$. Summed over the edges of P, or over one period if P is infinite, the right side counts every maximal run once, up to Π_P. So the terms $1/\rho_i$ sum to at most k_P. Number its maximal runs $I_0, I_1, \ldots$, starting anywhere if P is infinite, so that a generator of Π_P maps I_c to I_{c+k_P}. Pick an index $t_d \in I_d$ for every $d \equiv 0 \pmod{12}$. For $c \equiv 6 \pmod{12}$ with $c + 6 \leq k_P - 4$, let x_c be a vertex of an adhesion with index in I_c. By (R1), the adhesions with indices in I_c, $I_{c\pm3}$ and $I_{c\pm6}$ are pairwise disjoint. Every adhesion separates $\hat{G}$, so every path from x_c to $X_{t_{c\pm6}}$ has an inner vertex in an adhesion with index in $I_{c\pm3}$. Hence $N[x_c]$ consists of vertices owned by the hairs of the nodes p_i with $t_{c-6} < i \leq t_{c+6}$. For distinct c these nodes are distinct. If P is infinite, they lie in one period, since t_{c+6} lies before the left end of I_{k_P-2}, and thus before the left end of the translate I_{k_P} of I_0. There are at least $(k_P - 28)/12$ values of c. By (d) and the packing bound, $\sum_P k_P \leq 12\gamma_f + 28(4h + 2N_1)$,

over the core paths up to Π. Now let p be a node of a core path with $\lambda(p) \geq 6$. Its bag is a face, since thin edges have $\lambda \leq 1$ and vertex nodes have $\lambda = 0$, and by (ii) it is an isometric cycle. A shortest path in it between the two core adhesions of p contains at least $(\lambda(p) - 5)/3$ vertices at distance at least 3 from both adhesions and at pairwise distance at least 3. Their closed neighbourhoods consist of vertices owned by hair(p). By the packing bound, $\sum_p \lambda(p) \leq 3\gamma_f + 5\,|\{p : \lambda(p) > 0\}|$. A node p_{i+1} with $\lambda(p_{i+1}) > 0$ that is not an end of its core path has $X_i \cap X_{i+1} = \emptyset$, so a maximal run ends at i. So at most $k_P + 2$ such nodes lie on P, up to Π. Altogether,

$$\sum_e \mu(e) \leq \sum_P k_P + \sum_p \lambda(p) \leq 6 \sum_P k_P + 3\gamma_f + 10(4h + 2N_1) \leq 75\gamma_f + 178(4h + 2N_1).$$

So $\sum_{[e]} c(e) \leq 150\gamma_f/R + (1424/R + 12)h + (712/R + 8)N_1$. Take $\ell = R + 13$. For $R \geq 1500$, the bounds on h and N_1 give $\sum_{[e]} c(e) \leq 215\gamma_f/R$, so $\sum_v x_v \leq \gamma_f + 430\gamma_f/R$. Take $R = \lceil 1500/\varepsilon \rceil$. The windows W with $\hat{v} \in N[W]$, together with $N[W]$, lie within distance $6R + 2\ell + 10$ of $\hat{v}$. By (e) and the locality of K and μ, these windows, their weights and the problems $\mathrm{opt}^+(W)$ are determined by the ball of radius $9R + 5\ell + 10$ around $\hat{v}$. By (iv), v computes x_v in $9R + 6\ell + 10 = 15R + 88 \leq 22603/\varepsilon$ rounds.

Randomized version. Let $\ell = 6R + 5$. Cut every core edge at a branch node, and every edge e of a core path with $\mu(e) > R$. This splits the core paths into *segments*. Every edge e of a segment gets an exponential clock of rate $\mu(e)$, that is, the value $-\ln U/\mu(e)$ for a uniform value $U \in [0, 1]$ given by the binary expansion of a sequence of coins. There is one clock for each edge of D, drawn by the vertex with the smallest identifier in its adhesion, so the clocks are invariant under Π. Ties are broken by the identifier of the drawing vertex and then by the order in which it assigns its coins to its edges. For a node p of a segment, let $B(p)$ be the set of edges of the segment whose ends have ψ-distance at most R from p. If p has an edge in the segment, this edge lies in $B(p)$, and the *anchor* of p is the edge of $B(p)$ with the smallest clock. Cut every edge of a segment whose ends have different anchors. The clusters are the components of the remaining core with their hairs, and the components of T that miss K. All core nodes of a cluster lie within ψ-distance R of the ends of a common anchor. Since μ takes finitely many positive values, a ψ-interval of bounded length contains finitely many nodes, and every cluster is finite. As for the windows, (R3) shows that it has radius $O(R + \ell) = O(R)$. The clocks, the cuts and the hairs are invariant under Π, so Π permutes the clusters, and they partition the nodes of T. If a cluster C and gC share a node, then $gC = C$, and $g = 1$ as for the windows. So π is injective on the vertices owned by a cluster, and the projections of the clusters partition $V(G)$. Each projected cluster C outputs the lexicographically first smallest set in $N[C]$ that dominates C. The walk argument, applied to a lift of a boundary vertex of C, bounds its size by $|S \cap C| + \sum_{e \in \partial Q} |X_e|$ for a minimum dominating set S. So the expected size is at most $\gamma + 4\,\mathbb{E}[\#\text{cut edges of } D]$.

Let e_i and $e_{i+k} = ge_i$ with $g \neq 1$ and $k > 0$ be edges of one core path. By (iii), $d(X_i, X_{i+k}) > \ell - 1$, so $\sum_{j=i}^{i+k} \mu(e_j) > (\ell - 4)/2 > 3R$ by (R3). Let $e = pp'$ be an edge of a segment, with $\psi(p) < \psi(p')$. The edges in $B(p) \cup B(p')$ lie in a ψ-interval of length $\mu(e) + 2R \leq 3R$, so they are lifts of distinct edges of D, and their clocks are independent. In particular, ties have probability 0. The anchors of p and p' differ only if the smallest clock in $B(p) \cup B(p')$ lies in $B(p) \triangle B(p')$. This happens with probability $\mu(B(p) \triangle B(p'))/\mu(B(p) \cup B(p'))$, where μ of a set of edges is their total length. An edge f whose ends sit at ψ-positions $x < y$ lies in $B(q)$ exactly when $\psi(q) \in [y - R, x + R]$. If $f \in B(p) \setminus B(p')$, then $\psi(p) - R \leq x < \psi(p') - R$, and all edges of the segment between x and $\psi(p')$ lie in $B(p)$. So $\mu(B(p)) > R$ whenever $B(p) \not\subseteq B(p')$, and symmetrically for p'. Hence e is cut with probability at most $\mu(B(p) \triangle B(p'))/R$. Moreover, $f \in B(p) \triangle B(p')$ only if $y - R \in (\psi(p), \psi(p')]$ or $x + R \in [\psi(p), \psi(p'))$. This happens for at most two edges e of the segment. Summing over the orbits, we bound the expected number of anchor cuts in D by $2 \sum_f \mu(f)/R$, over the edges f of D

on core paths. At most $\sum_f \mu(f)/R$ of them have $\mu(f) > R$, and at most $6h + 4N_1$ core edges meet a branch node. With the bounds of the counting and $\ell = 6R + 5$, at most $242\gamma_f/R$ edges of D are cut in expectation for $R \geq 2200$, and the expected size is at most $\gamma + 968\gamma/R$. Take $R = \lceil 2200/\varepsilon \rceil$. Whether an edge of a segment is cut depends on μ and on the clocks within ψ-distance R of its ends. A cluster spans a ψ-interval of length at most $3R$. By (R3), (b) and (c), the clusters C with $\hat{v} \in N[C]$, and all edges whose clocks decide them, lie within distance $O(R + \ell)$ of $\hat{v}$. By (e), the locality of K and μ, and (iv), v computes its output in $O(R + \ell) = O(1/\varepsilon)$ rounds. □

The numerical checks are in `outer_frac.py` and `outer_cover.py`. The first works in G itself, on graphs whose holes are longer than all windows: rings, chains of holes and random outerplanar graphs of maximum degree 3, 4 and 6, and graphs with vertices of degree up to 320, namely fans, strips of alternating fans, Farey graphs, a hole with large fans glued to its outside, and random graphs that attach new faces preferentially at vertices of high degree. For $R \in \{4, 8, 16\}$, the windows have weight exactly 1 at every vertex, the averaged x dominates, and the excess $\sum_v x_v - \gamma_f$ is at most the repair bound. On these graphs that bound stays below $19\gamma_f/R$, against $430\gamma_f/R$ in the proof, and it does not grow with the maximum degree. Every window lies within distance $R + \ell$ of one of its vertices. For the randomized version, the expected number of long and anchor cuts stays below $1.1 \sum_f \mu(f)/R$, against $3 \sum_f \mu(f)/R$ in the proof.

The second script builds T and $\hat{G}$ and runs both versions on them as in the proof. It checks the intermediate claims one by one: the faces, the cycle rank of D, the separators and the packing behind the bound on h, (i)–(iii), (b)–(d), (R1)–(R3), the packing sets behind the bounds on N_1, $\sum_P k_P$ and $\sum_p \lambda(p)$, the bounds on $c(e)$ and on the diameters of the windows, and the repair bound for every window. For the randomized version, it computes the cut probabilities of the anchors exactly and samples the clusters. It runs on 166 graphs with up to 618 vertices, among them windows that wrap around short holes and contain several lifts of a vertex, holes of length exactly $\ell + 1$, vertices of degree up to 201, and random choices of the links of vertex nodes and of the ownership. No check fails. With $\ell = R + 13$, holes of length $\ell + 1$ and $R \leq 8$, the repair bound reaches $23.2\gamma_f/R$. When ℓ is much smaller than R, as in the tests of wrapping windows, the bound on h does not apply, and the repair bound reaches $124\gamma_f/R$. With $\ell = 6R + 5$ and $R \leq 3$, the expected number of cut edges of D stays below a third of the bound $3 \sum_f \mu(f)/R + 6h + 4N_1$ of the proof.

6 Formal verification

The Lean files and the scripts named in this paper are in the repository

`https://github.com/ssenge/MinimumDominatingSetOnPlanarGraphs`,

in the directories `lean/Op4/`, `sim/tight/` and `sim/lower/`. The Lean version is `v4.36.0-rc1`, and the Mathlib revision is pinned in `lean/lake-manifest.json`. Every checked theorem depends only on the standard axioms, and no proof is left open.

The file `RandomCentres.lean` formalizes Section 2, with the random centre set modelled exactly as independent coins. It checks Lemma 2 (`gain_eq_join_edges`), Lemma 3 (`vertex_bound`), Theorem 4 (`random_centres`), the identity between the arc sum and the edge weight, the value $f_3(3/8) = 1935/12288$ and the bound $f_3 \leq 63/400$, Lemma 5, and the bounds $6.4 < \ln 3/\ln(1/(1-f_3(3/8))) < 6.42$, $\ln 3/\ln(1/(1-f_3(1/2))) < 7$ and $25 < \kappa < 26$. The file `Orientation.lean` proves the direction of the orientation theorem of [8] that we use, by Hall's theorem: if every vertex set S spans at most $k|S|$ edges, there is an orientation with out-degree at most k (`orientation_of_density`). So Theorem 4 holds for every such graph (`random_centres_density`).

The file `Sharper.lean` checks Section 2.1. Theorem 10 (`sharper`) is proved in a discrete form of Lemma 7, by induction on the number of edges of positive weight: the smallest positive weight t is subtracted from every positive edge, and the gain splits as t times the join indicator plus the gain of the remaining weights. Also checked are Lemma 9 from (A) and (B) (`degree_lemma`), the certificates for $p = 3/8$ and $p = 1/2$ (`cert_three_eighths`, `cert_half`), the values of φ, and the bounds $5.88 < \kappa' < 5.89$ and $\kappa'_{p=1/2} < 6.7$. In this file, Lemma 8 enters as a hypothesis on every symmetric weighting whose support lies in that of w. The certificate of Lemma 9 is checked for every $p \in [21/64, 1]$ (`cert_range`), with h_1 and h_2 bounded as in the proof. So `sharper_range` gives Theorem 10 on the whole range, and `kappa_six_coins` gives $5.82 < \kappa' < 5.83$ for $p = 21/64$.

The file `General.lean` checks Section 2.2: Theorem 12 (`general`, for finitely many real thresholds, through the layer-cake core `sharper_core`), its degree inequality (`degree_lemma_gen`), and Corollary 13 (`aonly`, with the certificate `cert_aonly`). Here too, the facts of Lemma 11 enter as hypotheses.

The file `Density.lean` removes these hypotheses. It proves Lemmas 8 and 11 from two bounds on the support of w: every vertex set S spans at most $a|S|$ edges, and for disjoint S and T at most $b(|S| + |T|)$ edges join S to T (`facts_of_density`, `planar_facts`). Both bounds pass to every weighting whose support lies in that of w. Hence Theorem 10 for every $p \in [21/64, 1]$ (`sharper_range_density`) and Theorem 12 (`general_density`) need only these two bounds on w, and Theorem 4 needs only the first (`random_centres_densityA`). For planar graphs, $a = 3$ and $b = 2$ by Euler's formula, which we take as given.

The file `Surfaces.lean` checks every row of Table 2 with $e \geq 3$: the certificate (`cert_genus`e), the conditions $t \geq a_e$ and $u \geq 2a_e$ on its thresholds, the per-level bound under the density bounds a_e and b_e, obtained from `general_density` (`surface_genus`e), and the exponent bound in the last column (`kappa_genus`e). The densities a_e and b_e come from Euler's formula on the surface. The row $e \leq 2$ is Theorem 10.

The file `Asymptotic.lean` checks Proposition 6: the limit 8 for every $p(a)$ with $p \to 0$ and $ap \to \infty$ (`expRatio_tendsto_eight`), the limit $8(1 - p)$ for a fixed p (`expRatio_tendsto_const`), and the dyadic choice (`dyadic_p`). It also checks that with $p = 1/2$ the exponent is at most that of [1] for every $a \geq 1/2$ (`never_worse`).

The file `Table.lean` checks Table 1. It is generated from exact rational data by the script `lean/scripts/gen_table.py`. For every row it checks the three exponents as printed, that the printed p maximizes f_a among all $p = j/256$ (`best_dyadic_`a), and the improvement factor. It also checks the six-coin remark after Theorem 1: for every $p \leq 1$ with $\varphi(p) > 0$, the exponent for $p = 21/64$ is less than the exponent for p plus 10^{-3} (`six_coins_near_best`).

The file `Kkn.lean` checks Proposition 14. In a graph without triangles, every edge inside a star is a join edge (`internal_subset`). On $K_{k,n}$ with unit weights, for every $p \in [0, 1]$ and every choice among the centre neighbours, the expected number of internal edges is at most $(f_k(p) + 1/n)\,|E|$ (`kkn_internal`). So no analysis of the rule gives a per-level fraction above $f_k(p)$ on a class that contains all $K_{k,n}$ (`kkn_optimal`). For $k = 2$ the file also checks $\max_p f_2(p) = 1/(3\sqrt{3})$ and the exponent bound 5.13 of Section 4 (`kkn_planar`).

The file `Levels.lean` checks the step from the per-level factor to the number of rounds in Section 3. If $c_{t+1} \leq q\, c_t$ for every t and $x \geq 1$, then $L = \lceil \log x / \log(1/q) \rceil$ levels give $c_L \leq c_0/x$ and $3^L \leq 3x^\kappa$ with $\kappa = \ln 3 / \ln(1/q)$ (`levels`). This is how the factor q' enters the exponent of the round bound $T_{\mathcal{C}} + 9 \cdot 3^L$ of [1, Thm. 5.4]. The algorithm of [1] and the rest of its analysis are not formalized.

The files `Apollonian.lean`, `Stars.lean`, `Nest.lean` and `TriGrid.lean` check the other limits of Section 4. That the graphs used there are planar is taken as given.

- `Apollonian.lean` builds the Apollonian network. A face at level j is addressed by a face of K_4 and a word in $\{0,1,2\}^j$, and its child k replaces corner k by the inserted vertex. After ten steps, the vertices of K_4 have degree $3 \cdot 2^{10}$, a vertex inserted at level j has degree $3 \cdot 2^{9-j}$, and there are 118100 vertices and 354294 edges. For every p, the expected join gain with unit weights is at most $0.17402\,|E|$ (`apo_join_le`, by a polynomial certificate). At $p = 3/8$ it is $0.17161\,|E|$ to five decimals, and a per-level fraction of at most 0.17402 gives $\kappa > 5.74$.

- `Stars.lean` checks the square grid. In a graph without triangles and with maximum degree Δ, a star contraction internalizes at most $\Delta n/(\Delta+1)$ edges. On the $m \times m$ grid this is at most $4m^2/5$ of its $2m(m-1)$ edges (`grid_internal`), so $\kappa > 2.15$.

- `Nest.lean` checks Proposition 15. The graph has $n = 4k$ vertices, $3n-6$ edges and domination number k, and every partition into stars has at most $3k$ join edges (`nest_join`). Hence the fraction is at most $1/4$, and $\kappa > 3.818$.

- `TriGrid.lean` checks Proposition 16. A part sees which positions at offsets in $[-r,r]^2$ exist. This window contains its radius-r ball, and the part becomes a centre independently with a probability that depends only on the window and on m. The expected number of join edges on the $m \times m$ triangular grid is at most $0.61974\,m^2 + 4(r+1)m$ (`idfree_joins`), and the grid has at least $3m^2 - 12(r+1)m$ edges. So the fraction is at most 0.20658, and $\kappa > 4.74$.

The file `PathLower.lean` checks the counting in Theorem 17. Every fractional dominating set of P_{2R+M} with $M \geq 4$ that is constant on the middle and invariant under reflection has weight at least $(2R+M)/3+2/3$ (`path_excess`). For $L = 2r+6+(r \bmod 3)$, 3 divides L and $\gamma(P_L) = \gamma_f(P_L) = L/3$ (`path_gamma`). So the weight is at least $(1+2/L)\gamma_f(P_L)$, where $1+2/L \geq 1+1/(r+4)$ (`path_lower`), and ratio $1+\varepsilon$ forces $r \geq 1/\varepsilon - 4$ (`rounds_lower`).

The files `LocalPath.lean`, `LocalRandom.lean` and `LocalMeasure.lean` check the symmetrization over identifiers in an explicit model of r-round algorithms on P_L. The view of a vertex lists the identifiers and the degrees of the vertices at offsets $-r, \ldots, r$, with an empty entry for offsets outside the path. In the randomized case it also lists their coins. The coins are independent of each other and of the identifiers, and all have the same law on an arbitrary probability space, so real-valued randomness is included. A ball in a path has no preferred direction, so the algorithm is a function of the view that does not change when the view is read backwards. It may depend on L, r and the size of the identifier space. Suppose its output is a fractional dominating set for every injective identifier assignment and, in the randomized case, every outcome of the coins. In the randomized case the output at a vertex must also be integrable in the coins, which holds if it is measurable and bounded. Then for some assignment the (expected) weight is at least $(1+2/L)\gamma_f(P_L)$ (`local_path_lower`, `measure_path_lower`). Hence the ratio is at least $1+1/(r+4)$, and ratio $1+\varepsilon$ forces $r \geq 1/\varepsilon - 4$ (`local_rounds_lower`, `measure_rounds_lower`). As in the proof of Theorem 17, the identifier assignments and the coins are averaged together; permuting the coordinates of an independent and identically distributed vector of coins does not change its law. For coins with finitely many values, `LocalRandom.lean` gives an elementary version with finite sums.

The file `PathUpper.lean` checks the upper bound on paths in the same model. Every fractional dominating set of P_L has weight at least $\lceil L/3 \rceil$ (`path_lower_ceil`). For every $r \geq 0$ and $L \geq 1$, the algorithm described after the proof of Theorem 17 reads from the view only the distance to the nearer end, capped at $r+1$. So it does not change when the view is read backwards, and it may use L. For every injective identifier assignment it outputs a fractional dominating set of weight at most $(1+2/(2r+3))\,y(P_L)$ for every fractional dominating set y of P_L (`local_path_upper`). Hence for $0 < \varepsilon \leq 1$ some $r \leq 1/\varepsilon - 1/2$ gives ratio $1+\varepsilon$ (`local_rounds_upper`).

The file `Exact.lean` checks the first part of Corollary 18. Every vertex outputs whether it lies in a fixed minimum dominating set of its component, and the output is a minimum dominating set (`exact_min`). If neither G nor G' has a path on k vertices and the two graphs agree within distance k of v, the outputs at v are equal (`exact_local`). So k rounds suffice.

The file `Windows.lean` checks, for finite graphs, the steps of Theorems 19 and 20 that do not depend on the graph class: the packing bound (`packing`), the repair (1) (`repair`) and the averaging over windows (`window_average`). For the averaging, let the windows cover every vertex with total weight exactly 1. Let every window U come with a set B that meets $N[u]$ for every $u \in U$ with a neighbour outside U, and with a solution that dominates U and is optimal among the vectors supported on a given set that contains U and B. Then the weighted average of these solutions is a fractional dominating set of weight at most $y(V) + \sum_U \mathrm{wt}(U)\,|B|$ for every fractional dominating set y. This gives the first inequality of the averaging step in the proof of Theorem 19. In Theorem 20, `packing` applies as it is, and `repair`, on the finite graph induced by $N[W]$, gives the feasible solution of the repair step.

The files `ForestBasic.lean` to `ForestLocal.lean` check the deterministic part of the proof of Theorem 19 on every finite forest: (a) to (d), the counts of branch vertices, the cover by windows of total weight exactly 1 (`cover`, with `cover_count` on the core paths), the bounds $RL \le 3y(V)$ (`card_L_le`) and $|P| \le 3y(V) + 7L$ (`card_P_le`, with the counts of branch vertices), and the bound $(72/R)\,y(V)$ on the weighted boundary (`sum_wt_wB_le`). Lean lists the windows at an end e once for each of the $s(e)$ directions, and every component C that misses the core once for each of its vertices, with weight $1/|C|$. Instead of the optimum of least norm, the output averages one fixed optimal solution of each window problem, which exists by compactness (`exists_isOpt`). It is a fractional dominating set of weight at most $(1 + 72/R)\,y(V)$ for every fractional dominating set y (`forest_frac`). On any two forests in which the vertices at distance less than $7R$ from v have the same neighbours, the outputs at v are equal (`forest_local`). With $R = \lceil 72/\varepsilon \rceil$ for $0 < \varepsilon \le 1$, this gives ratio $1 + \varepsilon$ in $7R \le 511/\varepsilon$ rounds (`forest_frac_approx`). Here the vertex set is the identifier space, and the identifiers that are not nodes are isolated vertices; the bound then holds on the nodes (`forest_frac_on`).

The file `ForestRand.lean` checks the randomized part on every finite forest and for every $R \ge 1$. The input of [16] enters as a hypothesis: a set D_0 and stars d with $d(u) \in D_0 \cap N[u]$ and $d = \mathrm{id}$ on D_0. The values are independent with any atomless law on $\mathbb{R}$, for instance uniform on $[0, 1]$. Ties have probability 0 and are broken by a fixed index, and every cluster outputs a fixed minimum dominating set of V_W. The contraction H is a forest (`quot_acyclic`), adjacent vertices of a core path get different anchors with probability at most $2/(R+2)$ (`prob_anchor_ne`), and the clusters are those of the proof, and each lies within distance $2R$ in H of one vertex (`cl_shape`, `cluster_radius`). The output always dominates, and its expected size is at most $|D^*| + 16|D_0|/R$ for every dominating set D^* (`forest_rand`). Hence $|D_0| \le c\gamma$ and $R \ge 16c/\varepsilon$ give expected size at most $(1 + \varepsilon)\gamma$ (`forest_rand_approx`). Whether v is in the output depends only on the values within distance $15R+1$ of v (`randOut_local_G`). That the core and the clusters are computed from balls of radius $O(R)$, and the computation of D_0 in $O(1)$ rounds, are proved on paper only.

The file `Outer.lean` checks the steps of the proof of Theorem 20 that can be stated on their own, with the structure of the covering graph as hypotheses. A core path is modelled by adhesions X_i, each one vertex or one edge, such that for every vertex v the indices i with $v \in X_i$ form an interval. On such a path, Lean checks (R1), both bounds of (R2) and (R3) (`R1`, `R2_upper`, `R2_lower`, `R3`). Hence a path window of length at most R keeps its adhesions within distance $2R + 3$ of the first one (`window_radius`). It also checks $\sum_{\deg v \ge 3}(\deg v - 2) \le 2\,\mathrm{rk} + |\{v : \deg v = 1\}|$ for finite graphs (`branch_count`), the interval facts of the randomized version on a finite segment (`big_left`, `big_right`, `anchor_cuts`), the packing on an isometric cycle (`cycle_packing`), and the arithmetic

from the per-step bounds to the constants 75 and 178, 215 and 430, and 242 and 968 (`sum_mu_le`, `det_excess`, `rand_size`). The faces, the decomposition graph and its cycle rank, the covering graph with its group action, the core and the windows on it, the separation arguments behind the packing bounds, the averaging over Π-orbits with continuous weights and on infinite core paths, the race of the exponential clocks and the locality and round count are proved on paper only.

7 Open questions

- Can ratio $7+\varepsilon$ be achieved in poly$(1/\varepsilon)$ rounds with exponent close to 2, matching the $(8+\varepsilon)$ algorithm [23]? By [1], a deterministic $(1+\varepsilon)$-approximation of γ_f on planar graphs, with no degree bound, in such a number of rounds would suffice ([1, Open Problem 3]). By Corollary 18, the optimal number of rounds lies between $1/\varepsilon - 4$ and $O((c_0/\varepsilon)^{5.83})$, and for forests and for outerplanar graphs it is $\Theta(1/\varepsilon)$ (Theorems 19 and 20). Do planar graphs admit local windows of radius $\tilde{O}(1/\varepsilon)$ and weighted boundary $\varepsilon\gamma_f$ (Section 5.2)? Or is there a planar lower bound that is superlinear in $1/\varepsilon$?
- In a constant number of rounds, deterministic algorithms that output dominating sets cannot beat ratio 3 on paths [1, Thm. 8.7], while for fixed ε a $(1+\varepsilon)$-approximation on planar graphs takes $O(\log^* n)$ rounds [5]. How do the dependencies on ε and on n combine, already on paths?
- What is the worst case of the expected gain of the uniform rule on planar graphs? For the best p, it lies between 0.1718 (Theorem 10) and 0.1741 (Apollonian networks).
- Is there a rule in which the centre decisions depend on each other and that provably contracts more than $\varphi(p)$ per level? By Proposition 16, rules with independent coins cannot go below an exponent of 4.74 with an analysis that counts only join edges, and by Proposition 15 no such analysis of any star contraction goes below 3.81.

Use of AI

Claude Opus 5.5 (Anthropic) was used throughout this research: to explore approaches, to develop and check proofs, to write the simulation scripts and the Lean formalization, and to draft the text.

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